A Major-Project Report

on

IOT-BASED DOG DAYCARE ROBOTIC SYSTEM

Submitted in partial fulfillment for the Degree of B. Tech

in

Artificial Intelligence

by

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VIDYA JYOTHI INSTITUTE OF TECHNOLOGY

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2023-2024

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

CERTIFICATE

This is to certify that the project report entitled **"IOT Based Dog Daycare Robotic System"** Submitted by **KARTHIKESHWAR (20911A3522)**, **SHIVA NAINA (20911A3534), PREKSHA GUPTA (20911A3550), AVANCHA SANJANA (20911A3564)** to Vidya Jyothi Institute of Technology, Hyderabad, in partial fulfilment for the award of the degree of **B. Tech. in Artificial Intelligence** a *Bonafide* record of project work carried out by us under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree.

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Associate professor
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Signature of External Examiner

DECLARATION

We declare that this project report titled **"IOT Based Dog Daycare Robotic System"** Submitted in partial fulfilment of the degree of B. Tech in Artificial Intelligence is a record of original work carried out under the supervision of **Mr. A. Vijay Kumar**, and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgments have been made wherever the findings of others have been cited.

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ABSTRACT

Today automation technology is a trend that is easy to access, user friendly and easy to observe. Having pets can be a joy in our lives. Feeding pets is a major task and the purpose of this article is to automatically feed the pet when the owner is away from home. Over the past few years, many researchers have developed automatic feeding machine, but their functionality is limited to the home. This advanced machine can provide real-time food to the animal and monitor the animal's behavior. The IoT Based Automatic pet food feeder uses ESP-32 CAM camera module, Ultrasonic sensor, Motor control module, and consists of an interface with a DC servo motor, and other hardware equipment. A software code is dumped into the micro controller to perform operations of ultrasonic sensor, rotating motors. The web base application is developed by the PHP and HTML programming language and the real-time data link with the firebase database. The whole feeder system is controlled using a mobile phone. The user sends signals to the microcontroller using a mobile application through cloud. When the DC servo motor runs, the motor rotates the propeller which is in the feeding device. Using of ultrasonic sensor to get pet foods scale on real time. ESP32- cam to get video input to web application and could watch the behavior of the pet.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

In the area of wireless telecommunications, a new technology known as the Internet of Things (IoT) has received a huge interest in academia and industry over the last decades. These technologies have started to transform our lives in every way. The invention of the smartphone was the catalyst for the most significant transformation. Since then, we may have seen an increase in smart devices such as apps, games and tools with various functions. Those are acting by communicating with smartphones, which are carried by the majority of individuals nowadays [1]. As a result, we might refer to IoT as a new Internet concept in the twenty-first century [2]. New services that can be easily implemented around our real life can use the above concepts and technologies.

In the modern generation, almost everybody wants a pet because they believe that having a pet in the house reduces daily stress, boredom, and loneliness. However, in a developing country, citizen activity is at an all-time high, reducing the ability to provide necessary food and water for pets regularly. They are busy with various activities and have little time to keep an eye on their pets at home. In the current scenario, it is critical to provide food to pets in a variable quantity at a designated time. The pet food feeders' market is also related to intelligent technologies with the Internet of Things (IoT) and smartphones, which can provide convenient services with various aspects for pet owners. However, the current pet food feeders are restricted to simple functioning products such as the automatic feeder with timer and monitoring camera.

In this research work, we have proposed a new pet food feeder system with Real-time features that can feed the pets while the owners are absent from their homes, monitor their movement and status, and control move the robot through the owner's smart phones. And also, the system has a mobile feature that can follow the pet. The camera here can observe his activities. The proposed system is distinctive from others in terms that the proposed system is based on IoT technologies, which use many sensors and wireless communications. [17]

1.2 Problem statement

It is common to know that pet care burdens the pet owner. Any pet needs to be taken care of, and the owner needs to be there to take care of them. Some pet cannot control their diet and will eat as long as there's food for them. Other pets will eat a particular type of food. In other words, the owner cannot leave the pet alone. The problem occurs when the owner has to leave their pet for a particular time, and there's no one there to watch them. Therefore, to solve the problem, a system that can automatically feed the pet without the owner is needed to ensure that the pet stays healthy.

1.3 Objectives

1.Automated Supervision: Implementing robots equipped with cameras and sensors to monitor dogs' activities, ensuring their safety and well-being throughout the day. [3]

2.Real-time Tracking and Monitoring: Utilizing IoT devices such as GPS trackers and activity monitors to keep track of each dog's location, movement, and health parameters. This helps ensure that each dog receives appropriate care and attention.

3.Remote Interaction: Enabling owners to remotely interact with their pets through IoT-enabled devices such as cameras and interactive toys, providing reassurance and engagement for both the dog and the owner.

4.Health Monitoring: Integrating sensors to monitor vital signs such as heart rate, temperature, and activity levels, allowing for early detection of any health issues or abnormalities.

5.Automated Feeding and Watering: Implementing robotic feeding and watering systems controlled via IoT, ensuring that dogs receive their meals and hydration according to schedule and dietary requirements.

6.Behavioral Analysis: Using IoT devices to analyze behavioral patterns and preferences of dogs, enabling personalized care and enrichment activities tailored to individual needs.[4]

CHAPTER 2

LITERATURE SURVEY

2.1 RELATED WORK

There are many various types of pet food feeders on the market nowadays attempting to solve the problem of making sure that each animal has access to a healthy scale of food throughout the day, regardless of the owner's schedule.

"Smart Pet Feeder" has shown a prototype is offered to address the issue of supplying food and water for all types of pets when the owner is away from home, such as during a lockdown situation. The owner can provide food and water to their pet on their schedule by using this prototype. An ATMEGA32 microcontroller is used to control the proposed system. A servomotor has been added to the bowl to control the discharge of food through a PVC conduit. In the system the owner can assign the timing and quantity of food through a keyboard to supply food depending on the instantaneous weight of the food in the bowl.

According to Automatic pet feeder is a device that feeds food at predetermined intervals; all timings are preprogrammed in the microcontroller's program. There are two knobs and one dc motor on it. The first knob controls the feeding intervals, while the second knob controls the timing of the food outlet's opening and closing. The food outlet is opened and closed by a dc motor. A buzzer indicates the presence of food. There's a possibility that food will become stuck along the way because of the use of a vibrating dc motor.

Table 2.1 : Literature survey

S No.	NAME	PUBLICATIONS	DOI
1.	M Manoj	"AUTOMATIC PET FEEDER," International Journal of Advances in Science Engineering and Technology, vol. 3, no. 3, pp. 9–11, 2015	2015
2.	Friedemann Mattern and Christian	"From the Internet of Computers to the Internet of Things."	2010

	Floerkemeier	Institute for Pervasive Computing, 2010.	
3.	Soumallya Koley, Sneha Srimani, Debanjana Nandy, Pratik Pal, Samriddha Biswas, and Indranath Sarkar	Smart pet feeder," in Journal of Physics: Conference Series, Mar. 2021, vol. 1797, no. 1. doi: 10.1088/1742-6596/1797/1/012018.	2018
4.	[16] Brown, K., & White.	C. (2023). From Furry Friends to Tech Buddies: The Evolution of Dog Daycare with IoT-Driven Robotics. Robotics and Automation Letters, 9(1), 123-136.	2003
5.	M. Rohs and B. Gfeller	"Using Camera-Equipped Mo-bile Phones for Interacting with Real-World Object," Proceedings of Advances in Pervasive Computing	2004

2.2 Proposed System

New patent invention of Automatic Pet feeder is the device of the present invention relates to an automatic pet feeder, which has an integrated timer control module and pie-shaped bowl for feeding pets at preset programmed times. The device consists of a minimal number of parts.

The consumer can easily replace the parts in a matter of minutes in the event of a failure. The electronics device is housed in a single module which can be upgraded and customized to the requirements and needs of the consumer as time passes. Ease of cleaning, maintenance and replacement is facilitated, as the components are all made in plastic. The device operates with 3 AA batteries. The timer module can communicate with external devices so that a computer, gates, doors can control it, sensors such as infrared, proximity, motion etc. and other devices, such as a handheld remote.

"Smart Pet Monitoring and Feeding Based on Feedback Control System" was designed to produce the impact of continuous work in automation. It is an operating system approach controlled by the microcontroller system and based on the driver design and control principal technology. Its operation ensures that the food is released at precisely the right time, according to the schedule. The IoT can also be used to record and monitor the system.[6] The machine's basis of operation is that the food is stored in a silo with a screw conveyor inside that feeds the dog food. The use of an ultrasonic sensor, the camera to detect the dog's movement, and a loadcell for feedback control make up the machine. Every meal is weighted and scheduled for each mash and ensures that the dog gets the proper food at each meal. Second, the action is controlled by an IoT system operated by a mobile phone, and the system can be monitored at anytime. [5]

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

The first step of methodology, design the system architecture for the system. As pre the previous related works we design project works automatically and uses the devices mentioned in this chapter below. The operation of this project is shows in the Fig. 1. We used the ultrasonic sensor to get the pet foods scale, servo motor to open and close the value, L289N motor shield to control wheels and ESP32 CAM to get video input. Collect all the data using this ESP32 microcontroller. Then send this data to the cloud server. On another side, the client can control this device using a mobile phone or laptop like any internet-accessible device. The IoT architecture clearly show at Fig. 2.

The following components use for the hardware development.

3.2 Hardware Components

ESP32 CAM

ESP8266 WIFI Camera

ArduCAM now released the world's smallest low-cost ESP8266 WIFI IoT camera kit based on the ArduCAM-Mini-2MP-V2 and ArduCAM-ESP8266-Nano module. Users can implement a 2MP WIFI camera using HTTP or WebSocket protocol on ESP8266, and the camera can act as an AP, and mobile phone/PC can connect to the camera directly or act as a Station that connects to the home router. The kit can capture 2MP full-resolution JPEG still images, even stream low resolution at fairly frame rate video over a network or directly save to a local SD/TF card. The kit is suitable for portable application, it can be powered from micro-USB or using a battery and has built-in lithium battery charging circuits. The kit can also be used separately, it is almost identical to the standard alone ArduCAM-Mini-2MP camera and ESP8266-12F module.

Fig 3.1: Hardware Components

Features

- 2MP image sensor OV2640, support JPEG
- Standard FOV 60°stock lens
- I2C interface for the sensor configuration
- SPI interface for camera commands and data stream
- Onboard ES8266-12F module
- Build in Lithium battery recharging 3.7V/500mA max
- Build in SD/TF card socket
- Build in micro USB-Serial convertor
- Compatible with Arduino IDE
- Small form of factor

Specification

- Maximum resolution: 2MP
- Field of view: 60°
- Compression: JPEG/MJPEG
- Interface: SPI and I2C
- WIFI mode: *802.11* b/g/n software AP or Station mode
- Power consumption: 180mA @ 3.7V full running
- Mechanical size: 34mm(L) x 24mm(W) x 23(D)

Motor drive module

L293D

CP034

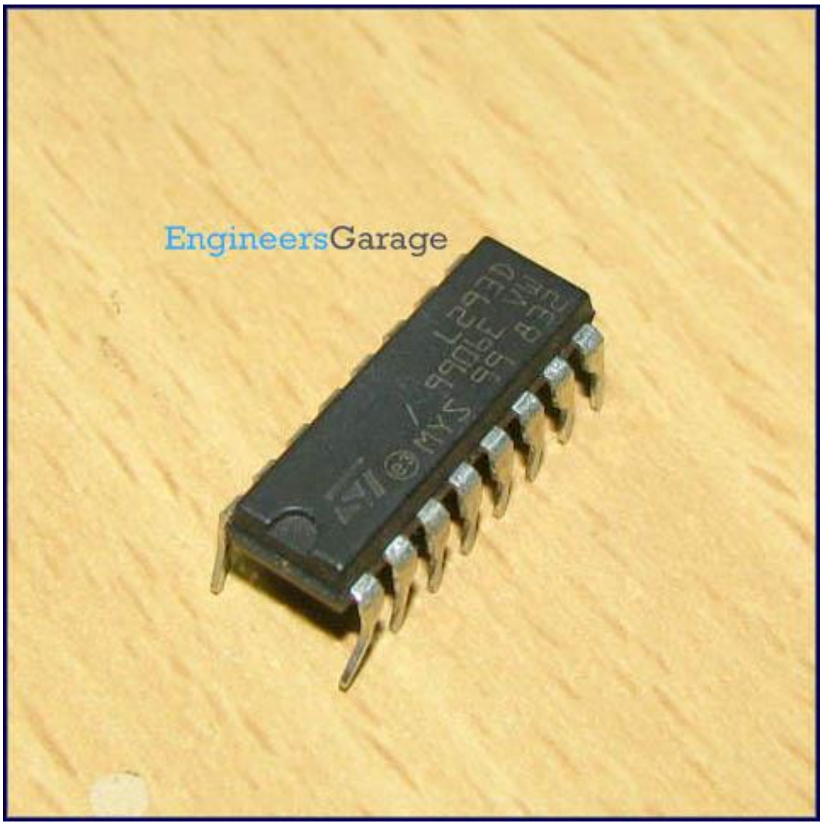


Fig 3.2: ESP32 CAM

A motor driver is an integrated circuit chip which is usually used to control motors in autonomous robots. Motor driver act as an interface between Arduino and the motors . The most commonly used motor driver IC's are from the L293 series such as L293D, L293NE, etc. These ICs are designed to control 2 DC motors simultaneously. L293D consist of two H-bridge. H-bridge is the simplest circuit for controlling a low current rated motor. We will be referring the motor driver IC as L293D only. L293D has 16 pins.

DataSheet:

L293D is a dual H-bridge motor driver integrated circuit (IC). Motor drivers act as current amplifiers since they take a low-current control signal and provide a higher-current signal. This higher current signal is used to drive the motors.

L293D contains two inbuilt H-bridge driver circuits. In its common mode of operation, two DC motors can be driven simultaneously, both in forward and reverse direction. The motor operations of two motors can be controlled by input logic at pins 2 & 7 and 10 & 15. Input logic 00 or 11 will stop the corresponding motor. Logic 01 and 10 will rotate it in clockwise and anticlockwise directions, respectively.

Enable pins 1 and 9 (corresponding to the two motors) must be high for motors to start operating. When an enable input is high, the associated driver gets enabled. As a result, the outputs become active and work in phase with their inputs. Similarly, when the enable input is low, that driver is disabled, and their outputs are off and in the high-impedance state.

Pin Diagram:

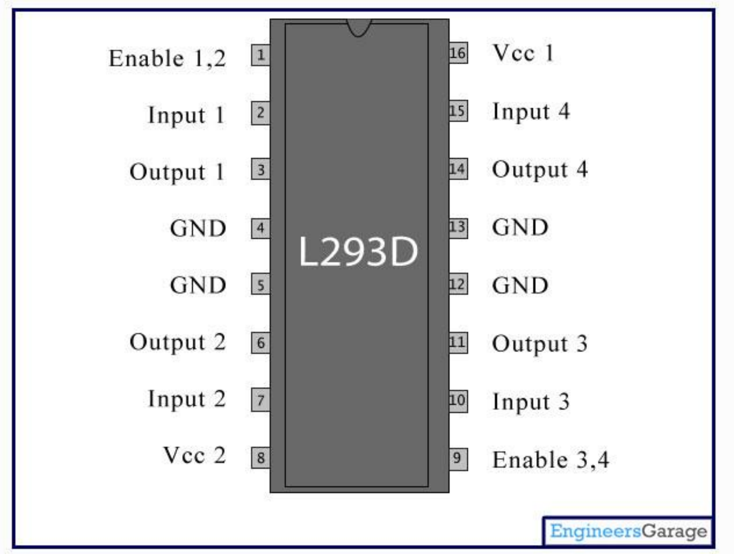


Fig 3.3: L293D Motor

Pin Description:

Pin No	Function	Name
1	Enable pin for Motor 1; active high	Enable 1,2
2	Input 1 for Motor 1	Input 1
3	Output 1 for Motor 1	Output 1
4	Ground (0V)	Ground
5	Ground (0V)	Ground
6	Output 2 for Motor 1	Output 2
7	Input 2 for Motor 1	Input 2
8	Supply voltage for Motors; 9-12V (up to 36V)	Vcc_2
9	Enable pin for Motor 2; active high	Enable 3,4
10	Input 1 for Motor 1	Input 3
11	Output 1 for Motor 1	Output 3
12	Ground (0V)	Ground
13	Ground (0V)	Ground
14	Output 2 for Motor 1	Output 4
15	Input2 for Motor 1	Input 4
16	Supply voltage; 5V (up to 36V)	Vcc_1

Servo motor

According to cost analysis, we use the MG995 metal gear servo motor. MG995 Metal Gear

Servo Motor is a standard high-speed servo that can rotate approximately 180 degrees (60 in each direction). Also, it provides 10kg/cm at 4.8V and 12kgcm at a voltage of 6V. The motor is a digital servo motor which receives and processes PWM signals better and faster. Its highly developed internal circuitry offers higher torque, holding power, and faster updates in response to outside forces. They are well sealed in a solid plastic container, making them water and dust-resistant, a quality that is highly beneficial for pet food feeder machines. The motor Operating Speed at no load (4.8V) is 20sec/ 60 deg. Operation Voltage is 4.8 to 7.2V. All gears are metal, which helps excellent holding power and accurate positioning.

Fig 3.4: Servo Motor

What is a Servo Motor? - Understanding the basics of Servo Motor Working

What is a Servo Motor? A **servo motor** is a type of motor that can rotate with great precision. Normally this type of

motor consists of a control circuit that provides feedback on the current position of the motor shaft, this feedback allows the servo motors to rotate with great precision. If you want to rotate an object at some specific angles or distance, then you use a servo motor. It is just made up of a simple motor which runs through a **servo mechanism**. If motor is powered by a DC power supply then it is called DC servo motor, and if it is AC-powered motor then it is called AC servo motor. For this tutorial, we will be discussing only about the **DC servo motor working**. Apart from these major classifications, there are many other types of servo motors based on the type of gear arrangement and operating characteristics. A servo motor usually comes with a gear arrangement that allows us to get a very high torque servo motor in small and lightweight packages.

Servo motors are rated in kg/cm (kilogram per centimeter) most hobby servo motors are rated at 3kg/cm or 6kg/cm or 12kg/cm. This kg/cm tells you how much weight your servo motor can lift at a particular distance. For example: A 6kg/cm Servo motor should be able to lift 6kg if the load is suspended 1cm away from the motors shaft, the greater the distance the lesser the weight carrying capacity. The position of a servo motor is decided by electrical pulse and its circuitry is placed beside the motor.

Servo Motor Working Mechanism

It consists of three parts:

1.Controlled device
2.Output sensor
3.Feedback system

It is a closed-loop system where it uses a positive feedback system to control motion and the final position of the shaft. Here the device is controlled by a feedback signal generated by comparing output signal and reference input signal.Here reference input signal is compared to the reference output signal and the third signal is produced by the feedback system. And this third signal acts as an input signal to the control the device. This signal is present as long as the feedback signal is generated or there is a difference between the reference input signal and reference output signal. So the main task of servomechanism is to maintain the output of a system at the desired value at presence of noises.

3.2.1 Servo Motor Working Principle

A servo consists of a Motor (DC or AC), a potentiometer, gear assembly, and a controlling circuit. First of all, we use gear assembly to reduce RPM and to increase torque of the motor. Say at initial position of servo motor shaft, the position of the potentiometer knob is such that there is no electrical signal generated at the output port of the potentiometer. Now an electrical signal is given to another input terminal of the error detector amplifier. Now the difference between these two signals, one comes from the potentiometer and another comes from other sources, will be processed in a feedback mechanism and output will be provided in terms of error signal. This error signal acts as the input for motor and motor starts rotating. Now motor shaft is connected with the potentiometer and as the motor rotates so the potentiometer and it will generate a signal. So as the potentiometer’s angular position changes, its output feedback signal changes. After sometime the position of potentiometer reaches at a position that the output of potentiometer is same as external signal provided. At this condition, there will be no output signal from the amplifier to the motor input as there is no difference between external applied signal and the signal generated at potentiometer, and in this situation motor stops rotating.

3.2.2 Interfacing Servo Motors with Microcontrollers:

Interfacing hobby Servo motors like s90 servo motor with MCU is very easy. **Servos have three wires coming out of them**. Out of which two will be used for Supply (positive and negative) and one will be used for the signal that is to be sent from the MCU. An **MG995 Metal Gear Servo Motor** which is most commonly used for RC cars humanoid bots etc. The picture of MG995 is shown below:

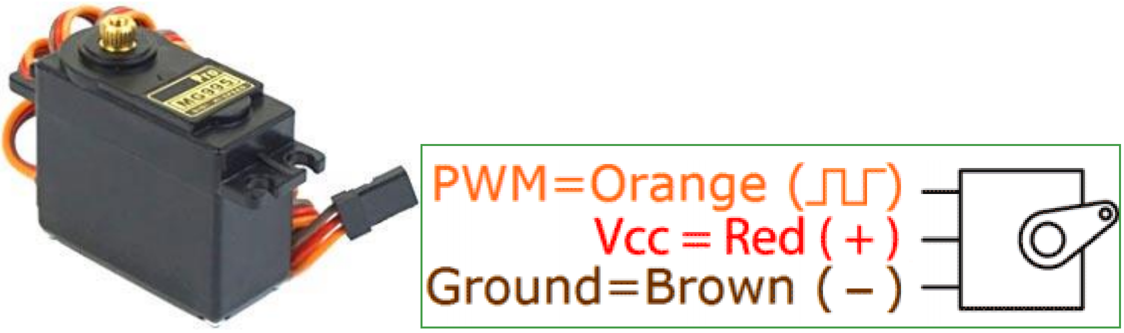


Fig 3.5: Interfacing Servo Motors with Microcontrollers:

The color coding of your servo motor might differ hence check for your respective datasheet.

All servo motors work directly with your +5V supply rails but we have to be careful on the amount of current the motor would consume if you are planning to use more than two servo motors a proper servo shield should be designed.

Controlling Servo Motor:

All motors have three wires coming out of them. Out of which two will be used for Supply (positive and negative) and one will be used for the signal that is to be sent from the MCU.

Servo motor is controlled by PWM (Pulse with Modulation) which is provided by the control wires. There is a minimum pulse, a maximum pulse and a repetition rate. Servo motor can turn 90 degree from either direction form its neutral position. The servo motor expects to see a pulse every 20 milliseconds (ms) and the length of the pulse will determine how far the motor turns. For example, a 1.5ms pulse will make the motor turn to the 90° position, such as if pulse is shorter than 1.5ms shaft moves to 0° and if it is longer than 1.5ms than it will turn the servo to 180°.

Servo motor works on **PWM (Pulse width modulation)** principle, means its angle of rotation is controlled by the duration of applied pulse to its Control PIN. Basically servo motor is made up of **DC motor which is controlled by a variable resistor (potetiometer) and some gears**.

High speed force of DC motor is converted into torque by Gears. We know that WORK= FORCE X DISTANCE, in DC motor Force is less and distance (speed) is high and in Servo, force is High and distance is less. The potentiometer is connected to the output shaft of the Servo, to calculate the angle and stop the DC motor on the required angle.

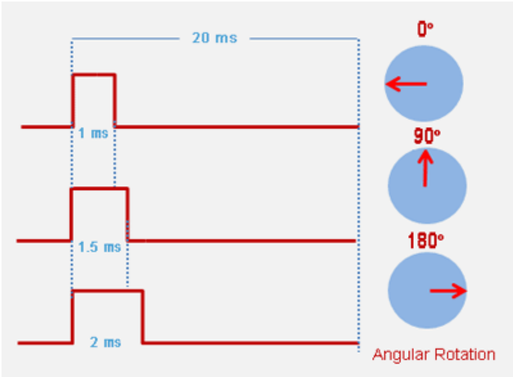


Fig 3.6 :Controlling Servo Motor

Servo motor can be rotated from 0 to 180 degrees, but it can go up to 210 degrees, depending on the manufacturing. This degree of rotation can be controlled by applying the **Electrical Pulse** of proper width, to its Control pin. Servo checks the pulse in every 20 milliseconds. The pulse of 1 ms (1 millisecond) width can rotate the servo to 0 degrees, 1.5ms can rotate to 90 degrees (neutral position) and 2 ms pulse can rotate it to 180 degree.

All servo motors work directly with your +5V supply rails but we have to be careful about the amount of current the motor would consume if you are planning to use more than two servo motors a proper servo shield should be designed.

Metel Gear motor:

Small brushed DC gear motors can deliver much power for their size. As per the literature review, we selected the brushed gear motor. The selected gear motor is a 12V brushed DC motor with long-life carbon brushes combined with a 32:1 metal spur gearbox. The D shaped output shaft is 4 mm in diameter and extends 18 mm from the face plate of the gearbox on the gearmotor, which has a cylindrical shape and a 20 mm diameter. It's no lord condition the speed is 450 RPM.

Ultrasonic Sensor:

HC-SR04 Ultrasonic Sensor Works & Interface It With Arduino

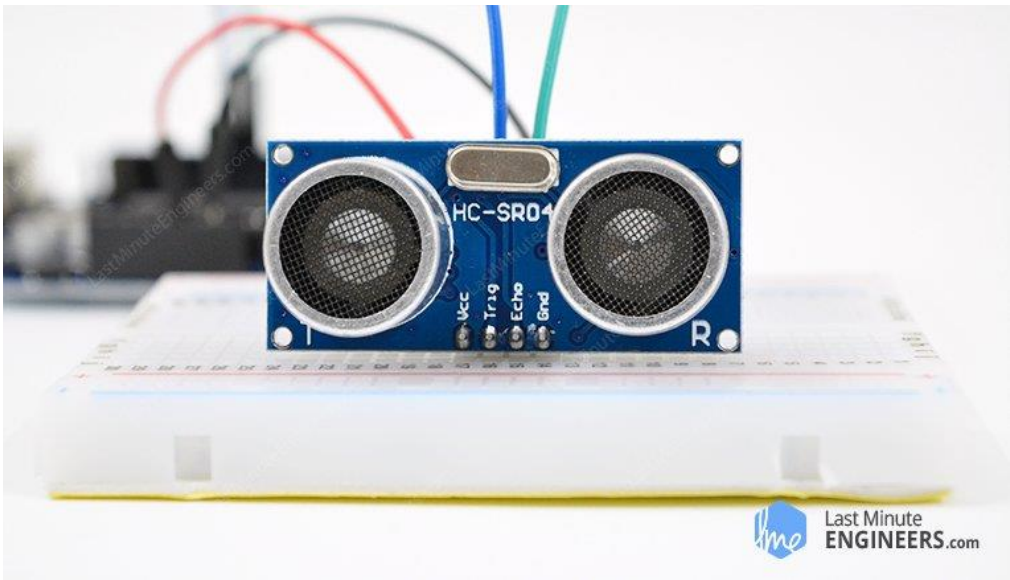
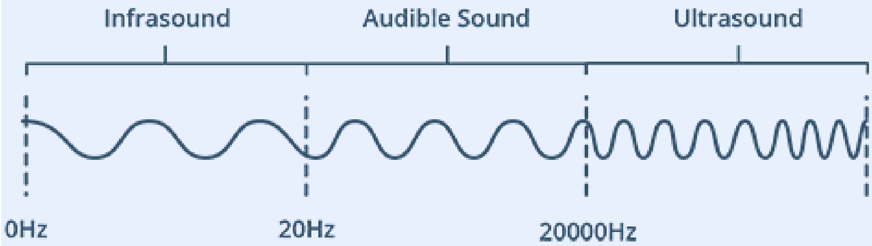


Fig 3.7: Ultrasonic Sensor

Give your next Arduino project bat-powers with a HC-SR04 Ultrasonic Distance Sensor that can report the range of objects up to 13 feet away. Which is really good information to have if you're trying to avoid your robot driving into a wall! They are low power (suitable for battery powered devices), inexpensive, easy to interface with and are insanely popular among hobbyists. And as a bonus it even looks cool, like a pair of Wall-E Robot eyes for your latest robotic invention!

What is Ultrasound?

Ultrasound is high-pitched sound waves with frequencies higher than the audible limit of human hearing.



Human ears can hear sound waves that vibrate in the range from about 20 times a second (a deep rumbling noise) to about 20,000 times a second (a high-pitched whistling). However, ultrasound has a frequency of over 20,000 Hz and is therefore inaudible to humans.

3.2.3 HC-SR04 Hardware Overview

At its core, the HC-SR04 Ultrasonic distance sensor consists of two ultrasonic transducers. The one acts as a transmitter which converts electrical signal into 40 KHz ultrasonic sound pulses. The receiver listens for the transmitted pulses. If it receives them it produces an output pulse whose width can be used to determine the distance the pulse travelled. As simple as pie!

The sensor is small, easy to use in any robotics project and offers excellent non-contact range detection between 2 cm to 400 cm (that's about an inch to 13 feet) with an accuracy of 3mm. Since it operates on 5 volts, it can be hooked directly to an Arduino or any other 5V logic microcontrollers.

Here are complete specifications:

Operating Voltage	DC 5V
Operating Current	15mA
Operating Frequency	40KHz
Max Range	4m
Min Range	2cm
Ranging Accuracy	3mm
Measuring Angle	15 degree
Trigger Input Signal	10µS TTL pulse
Dimension	45 x 20 x 15mm

HC-SR04 Ultrasonic Sensor Pinout

Let's take a look at its Pinout.

Sensor transmits a sonic burst of eight pulses at 40 KHz. This 8-pulse pattern makes the "ultrasonic signature" from the device unique, allowing the receiver to differentiate the transmitted pattern from the ambient ultrasonic noise.

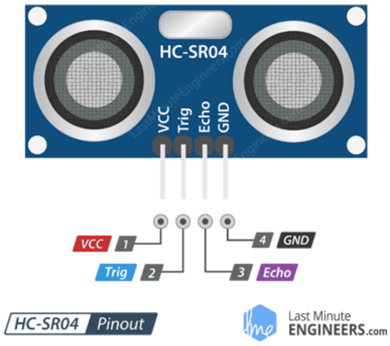


Fig 3.8: HC-SR04 Ultrasonic Sensor Pinout

VCC is the power supply for HC-SR04 Ultrasonic distance sensor which we connect the 5V pin on the Arduino.

Trig (Trigger) pin is used to trigger the ultrasonic sound pulses.

Echo pin produces a pulse when the reflected signal is received. The length of the pulse is proportional to the time it took for the transmitted signal to be detected.

GND should be connected to the ground of Arduino.

How Does HC-SR04 Ultrasonic Distance Sensor Work?

It all starts, when a pulse of at least 10 µS (10 microseconds) in duration is applied to the Trigger pin. In response to that the sensor transmits a sonic burst of eight pulses at 40 KHz. This 8-pulse pattern makes the "ultrasonic signature" from the device unique, allowing the receiver to differentiate the transmitted pattern from the ambient ultrasonic noise.

The eight ultrasonic pulses travel through the air away from the transmitter. Meanwhile the Echo pin goes HIGH to start forming the beginning of the echo-back signal.

In case, If those pulses are not reflected back then the Echo signal will timeout after 38 mS (38 milliseconds) and return low. Thus a 38 mS pulse indicates no obstruction within the range of the sensor. [19] Martinez, J., & Hernandez, M. (2023). Bridging the Gap: Exploring the Impact of IoT-Enabled Robotics on Canine Behavior in Daycare Environments. Robotics and Autonomous Systems, 98, 210-225.

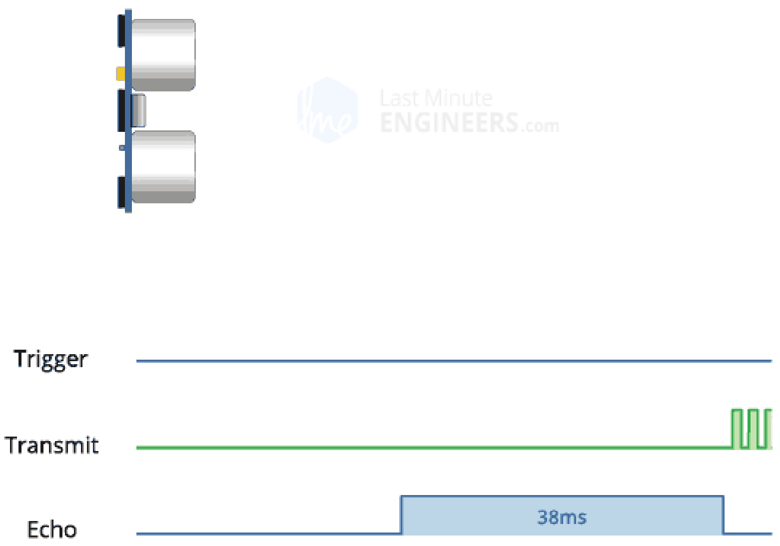


Fig 3.9: HC-SR04 Ultrasonic Distance Sensor Work

If those pulses are reflected back the Echo pin goes low as soon as the signal is received. This produces a pulse whose width varies between 150 µS to 25 mS, depending upon the time it took for the signal to be received.

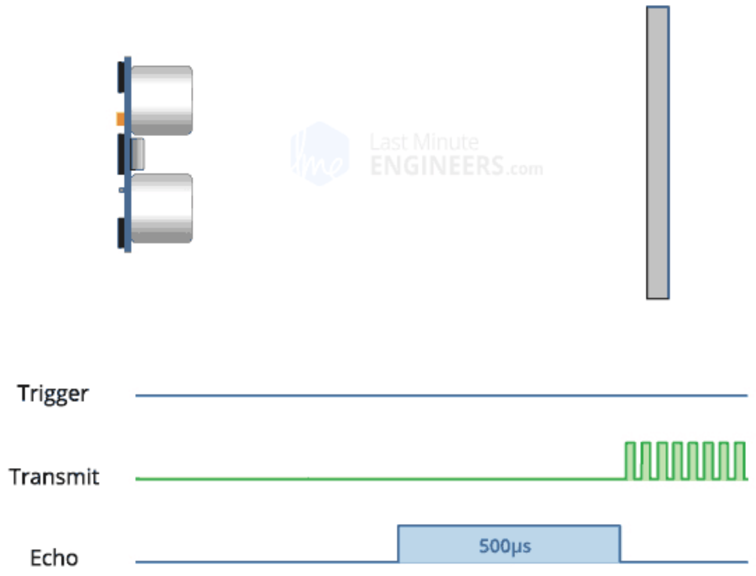


Fig 3.10:HC-SR04 Ultrasonic Distance Sensor Work

The width of the received pulse is then used to calculate the distance to the reflected object. This can be worked out using simple distance-speed-time equation, we learned in High school. In case you forgot, an easy way to remember the distance, speed and time equations is to put the letters into a triangle.

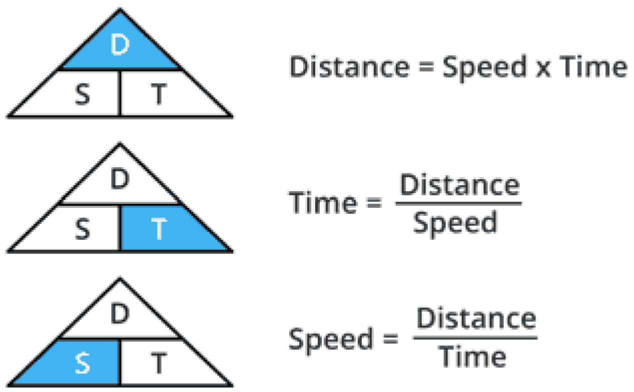


Fig 3.11: The distance, speed and time equations

Let's take an example to make it more clear. Suppose we have an object in front of the sensor

at an unknown distance and we received a pulse of width 500 µS on the Echo pin. Now let's calculate how far the object from the sensor is. We will use the below equation.

Distance = Speed x Time

Here, we have the value of Time i.e. 500 µs and we know the speed. What speed do we have?

The speed of sound, of course! Its 340 m/s. We have to convert the speed of sound into cm/µs in order to calculate the distance. A quick Google search for "speed of sound in centimeters per microsecond" will say that it is 0.034 cm/µs. You could do the math, but searching it is easier. Anyway, with that information, we can calculate the distance!

Distance = 0.034 cm/µs x 500 µs

But this is not done! Remember that the pulse indicates the time it took for the signal to be sent out and reflected back so to get the distance so, you'll need to divide your result in half.

Distance = (0.034 cm/µs x 500 µs) / 2

Distance = 8.5 cm

So, now we know that the object is 8.5 centimeters away from the sensor.

Wiring – Connecting HC-SR04 to Arduino Uno

Now that we have a complete understanding of how HC-SR04 ultrasonic distance sensor works, we can begin hooking it up to our Arduino!

Connecting the HC-SR04 to the Arduino is pretty easy. Start by placing the sensor on to your breadboard. Connect VCC pin to the 5V pin on the Arduino and connect GND pin to the Ground pin on the Arduino.When you're done you should have something that looks similar to the illustration shown below.

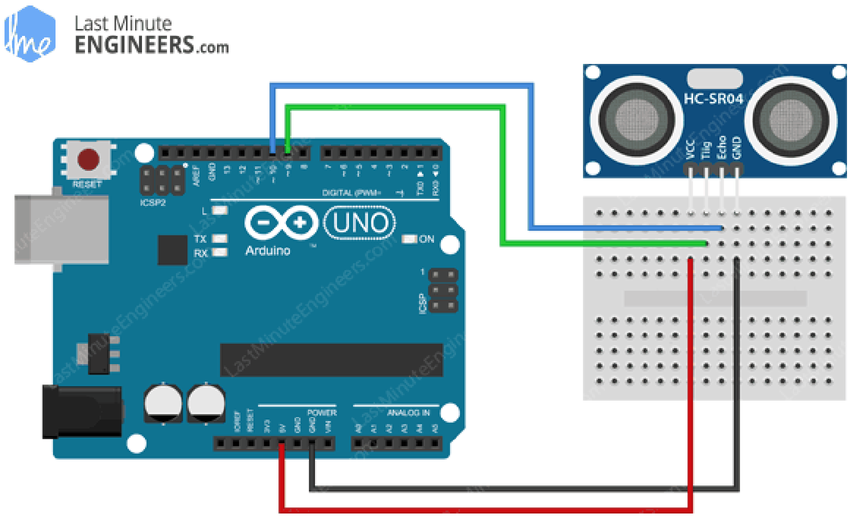


Fig 3.12: Wiring HC-SR04 Ultrasonic Sensor to Arduino UNO – Normal Mode

So now that we've hooked up our ultrasonic distance sensor it's time to write some code and test it out.

3.3 Software Components

Arduino Code – Using NewPing Library

Instead of triggering the ultrasonic sensor and measuring the received signal pulse width manually, we will use a special library. There are quite a few of them available, the most versatile is one called "NewPing".

Download the library first, by visiting the Bitbucket repo or, just click this button to download the zip:

To install it, open the Arduino IDE, go to Sketch > Include Library > Add .ZIP Library, and then select the NewPing ZIP file that you just downloaded. If you need more details on installing a library, visit this Installing an Arduino Library tutorial.

The NewPing library is quite advanced and it considerably improves upon the accuracy of our original sketch. It also supports up to 15 ultrasonic sensors at once and it can directly output in centimetres, inches or time duration.

Here is our sketch rewritten to use the NewPing library:

// This uses Serial Monitor to display Range Finder distance readings

3.4 Code

```
// Include NewPing Library
#include "NewPing.h"
// Hook up HC-SR04 with Trig to Arduino Pin 9, Echo to Arduino pin 10
#define TRIGGER_PIN 9
#define ECHO_PIN 10
// Maximum distance we want to ping for (in centimeters).
#define MAX_DISTANCE 400
// NewPing setup of pins and maximum distance.
NewPing sonar(TRIGGER_PIN, ECHO_PIN, MAX_DISTANCE);
float duration, distance;
void setup()
{
 Serial.begin(9600);
}
void loop()
{
 // Send ping, get distance in cm
 distance = sonar.ping_cm();

 // Send results to Serial Monitor
 Serial.print("Distance = ");
 if (distance >= 400 || distance <= 2)
 {
		Serial.println("Out of range");
 }
 else
 {
		Serial.print(distance);
```

```
        Serial.println(" cm");
 }
 delay(500);
}
```

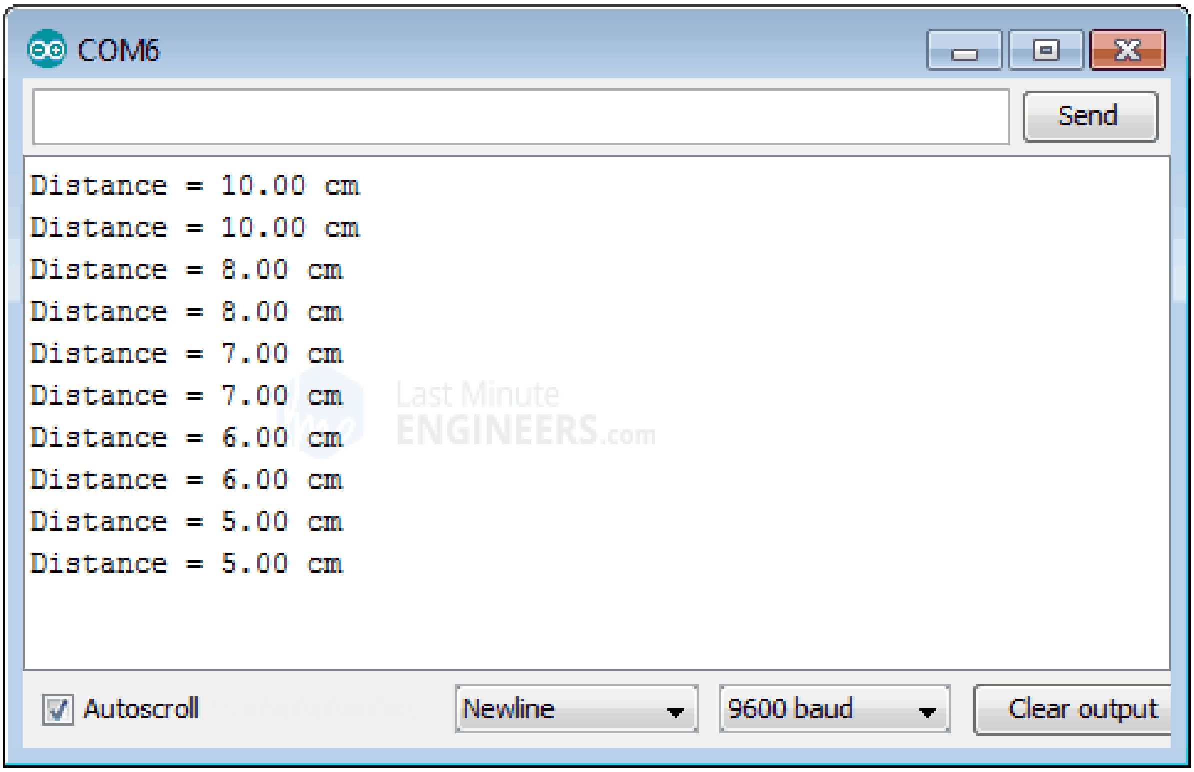


Fig3.13: Output on Serial Monitor

Code Explanation:

The above sketch is simple and works well but it only has a resolution down to one centimeter. If you want to bring back the decimal point values you can use New Ping in duration mode instead of in distance mode. You need to replace this line

```
// Send ping, get distance in cm
distance = sonar.ping_cm();
```

with below lines

```
duration = sonar.ping();
distance = (duration / 2) * 0.0343;
```

To improve the accuracy of your HC-SR04 to the next level, there's another function in NewPing library called "iterations". To iterate means to go over something more than once, and that's precisely what the iteration mode does. It takes many duration measurements instead of just one, throws away any invalid readings and then averages the remaining ones. By default it takes 5 readings but you can actually specify as many as you wish.

int iterations = 5;
duration = sonar.ping_median(iterations);
Arduino Project

Contactless Distance Finder

Let's create a quick project to demonstrate how a simple ultrasonic sensor can be turned into a Sophisticated Contactless Distance Finder. In this project we will use a 16×2 Character LCD to display a horizontal bar to graphically represent distance to the object with the value on the bottom line.

In case you are not familiar with 16×2 character LCDs, consider reading (at least skimming) below tutorial.

Interfacing 16×2 Character LCD Module with Arduino:
Want your Arduino projects to display status messages or sensor readings? Then these LCD displays might be the perfect fit. They are extremely common and...Next, we need to make connections to the LCD as shown below. [9] G. Kortuen, F. Kawsar, D. Fitton and V. Sundramoorthy, "Smart Objects as Building Blocks for the Internet of Things," IEEE Internet Computing, Vol. 14, No. 1, 2010, pp. 44-51. doi:10.1109/MIC.2009.143

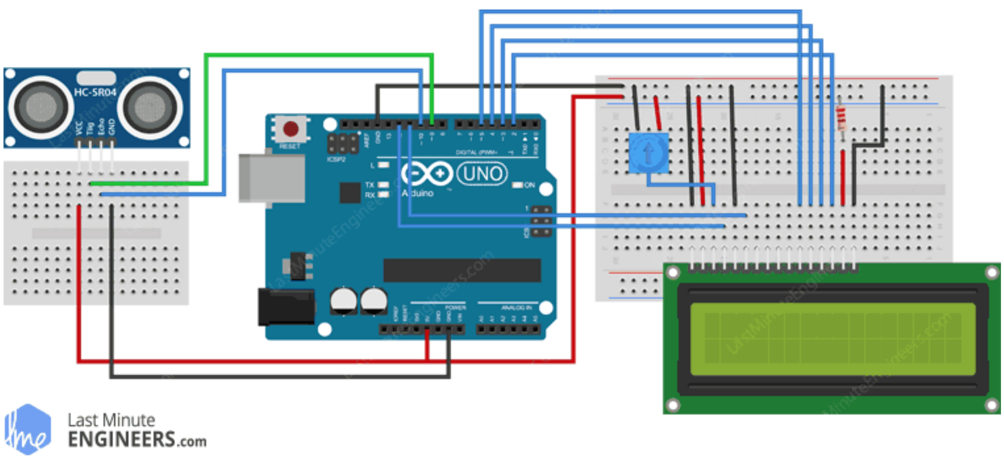


Fig 3.14: Wiring HC-SR04 Ultrasonic Sensor and 16×2 LCD to Arduino UNO

Before we get to uploading code and playing with the sensor, we need to install the library called LCDBarGraph. This library helps draw horizontal bargraph on the LCD, where the length of the bar is proportional to the values provided.

Download the library first, by visiting the Arduino Playground or, just click this button to download the zip:

Once you install the library, try the below sketch out.

```
// includes the LiquidCrystal Library
#include <LiquidCrystal.h>

// includes the LcdBarGraph Library
#include <LcdBarGraph.h>

// Maximum distance we want to ping for (in centimeters).
#define max_distance 200

// Creates an LCD object. Parameters: (rs, enable, d4, d5, d6, d7)
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

LcdBarGraph lbg(&lcd, 16, 0, 1); // Creates an LCD Bargraph object.
const int trigPin = 9;
const int echoPin = 10;
long duration;
int distance;

void setup()
{
 lcd.begin(16,2); // Initializes the interface to the LCD screen

 pinMode(trigPin, OUTPUT);
 pinMode(echoPin, INPUT);
}

void loop()
```

```
{
 // Write a pulse to the HC-SR04 Trigger Pin
 digitalWrite(trigPin, LOW);
 delayMicroseconds(2);
 digitalWrite(trigPin, HIGH);
 delayMicroseconds(10);
 digitalWrite(trigPin, LOW);

 // Measure the response from the HC-SR04 Echo Pin
 duration = pulseIn(echoPin, HIGH);

 // Determine distance from duration
 // Use 343 metres per second as speed of sound
 distance= duration*0.034/2;

 // Prints "Distance: <value>" on the first line of the LCD
 lcd.setCursor(0,0);
 lcd.print("Distance: ");
 lcd.print(distance);
 lcd.print(" cm");

 // Draws bargraph on the second line of the LCD
 lcd.setCursor(0,1);
 lbg.drawValue(distance, max_distance);
 delay(500);
}
```

This is how the output looks like.

fig 3.15: paragraph Output on 16×2 Character LCD

3.4.1 Code Explanation:

First you need to set up the LiquidCrystal library as usual. After this you can create LcdBarGraph instance with the just created LiquidCrystal instance. You should pass the reference of the LiquidCrystal to the constructor of the LcdBarGraph.

The constructor of the LcdBarGraph takes three more parameters. Second one is Number of character columns in the LCD (In our case that's 16). The last two parameters are optional and allow custom positioning of the bar.

```
// creating bargraph instance
LcdBarGraph lbg(&lcd, 16, 0, 1);
```

Now once we calculate the distance from the sensor, we can use drawValue(value, maxValue) function to display the bargraph. This draws a bargraph with a value between 0 and maxValue.

```
//display bargraph

lbg.drawValue(distance, max_distance);
```

3.4.2 Interfacing HC-SR04 with 3-Wire Mode

3-Wire Mode is something you only require one connection to a single Arduino digital I/O pin instead of two. If you don't know, there are many ultrasonic sensors out there that only operate in 3-Wire Mode like awesome parallax ping))) sensor.

In 3-Wire mode the single I/O pin is used as both an input and as an output. This is possible because there is never a time when both the input and output are being used. By eliminating one I/O pin requirement we can save a connection to our Arduino and use it for something else. It also is useful when using a chip like the ATtiny85 which has a limited number of I/O pins.

Here's how you can hook the HC-SR04 sensor up to the Arduino using 3-Wire mode.

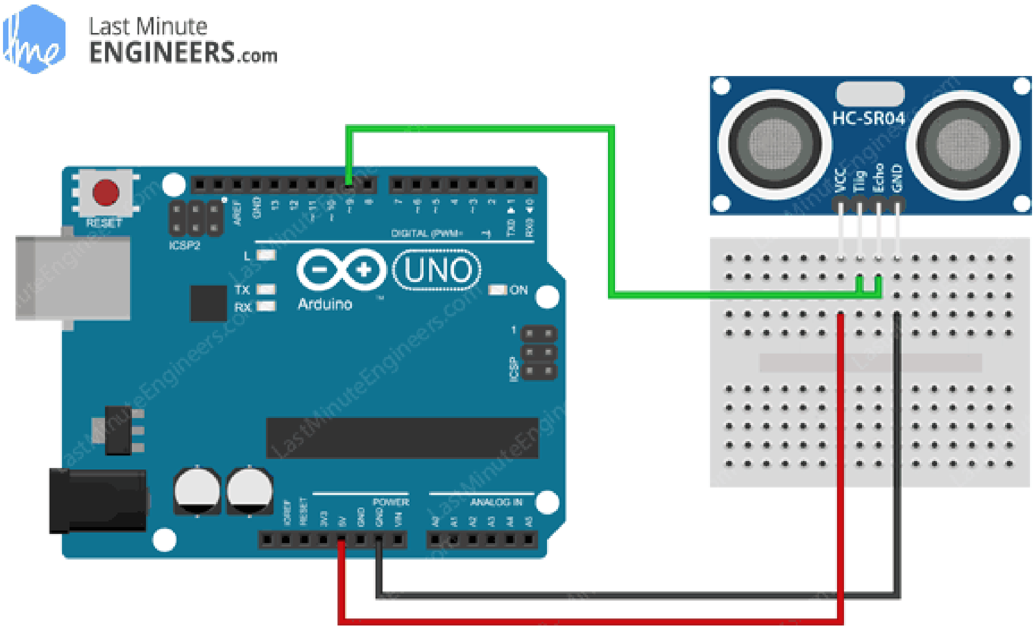


Fig 3.16: Wiring HC-SR04 Ultrasonic Sensor to Arduino UNO – 3 Wire Mode

As you can see all you need to do is, connect both the trigger and echo to Arduino pin 9.Note that the only difference you need to do in the sketch is to define pin 9 for both the Trigger and Echo pin values. The rest of the sketch is identical.

#define TRIGGER_PIN 9 // Trigger and Echo both on pin 9

#define ECHO_PIN 9

What are the limitations?

In terms of accuracy and overall usefulness, HC-SR04 ultrasonic distance sensor is really great, especially compared to other low-cost distance detection sensors. That doesn't mean that the HC-SR04 sensor is capable of measuring "everything". Following diagrams shows a few situations that the HC-SR04 is not designed to measure:

a) The distance between the sensor and the object/obstacle is more than 13 feet.

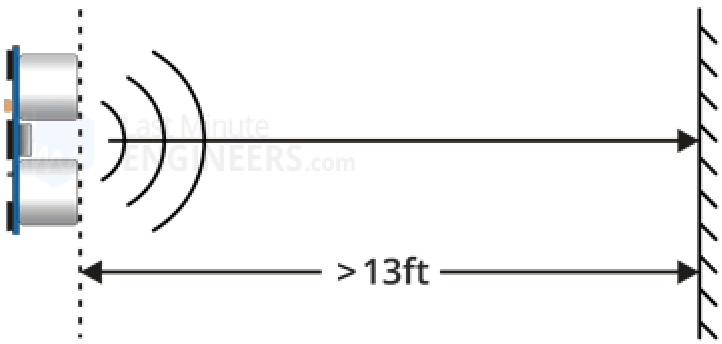


Fig 3.17: The distance between the sensor and the object

b) The object has its reflective surface at a shallow angle so that sound will not be reflected back towards the sensor.

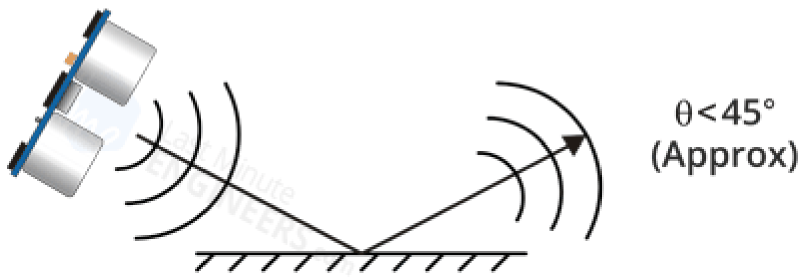


Fig 3.18: reflective surface

c) The object is too small to reflect enough sound back to the sensor. In addition, if your HC-SR04 sensor is mounted low on your device, you may detect sound reflecting off of the floor.

Fig 3.19: object is too small to reflect

d) While experimenting with the sensor, we discovered that some objects with soft, irregular surfaces (such as stuffed animals) absorb rather than reflect sound and therefore can be difficult for the HC-SR04 sensor to detect.

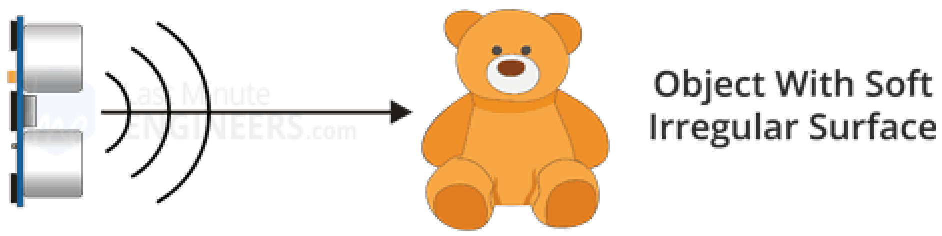


Fig 3.20: object with soft irregular surface

Effect of Temperature on Distance Measurement

Though the HC-SR04 is reasonably accurate for most of our projects such as intruder detection or proximity alarms; But there are times you might want to design a device that is to be used outdoors or in an unusually hot or cold environment.

If this is the case, you might want to take into account the fact that the speed of sound in air varies with temperature, air pressure and humidity.

Since the speed of sound factors into our HC-SR04 distance calculation this could affect our readings. If the temperature (°C) and Humidity is already known, consider the below formula:

Speed of sound m/s = 331.4 + (0.606 * Temp) + (0.0124 * Humidity)

Li-Po Battery Pack:

This pet feeder robot uses the Protek R/C Li-Poly 2300mAh receiver battery pack with ultra-thick 20awg silicone wire . This ultra-light Li-Poly battery is made to power the electronics of 1/10th and 1/8th scale nitro race cars. The primary advantage of using a Li-Poly receiver battery is that the battery voltage to your receiver and servos will remain constant during the race with the use of a voltage regulator. The process of pet food feeder and there will be procedures and check the machine operator with a working plan as a flow chart.

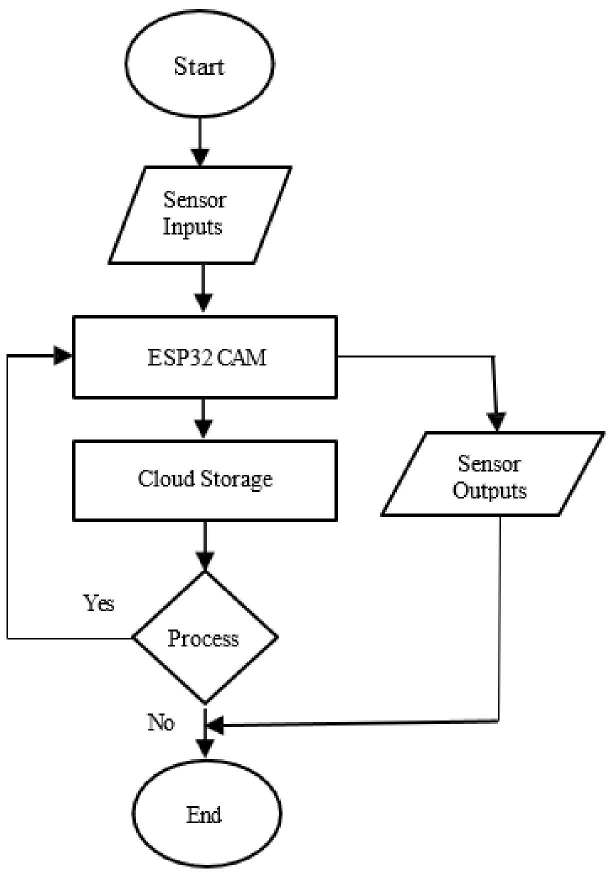


Fig 3.21: Flow Chart

3.5 Hardware Development

The whole operation collects data through the sensors and actuators. Which gathers all information regarding pets with the sensor implemented in the Smart Pet food feeder System. As shown in figure 8, the automatic feeder was implemented with a 3-D printer. Firstly, we implemented the food container and dispending using the 3D printer. The selected 3D printer material is High-density polyethene (HDPE). HDPE is a thermoplastic polymer produced from the monomer ethylene. Thermoplastic is a polymer that becomes plastic and flexible upon heating and hardens upon cooling. HDPE is made up of a string of ethylene molecules, which gives it the ‘poly’ in polyethene. It is well-known and liked for being strong, lightweight and easy to shape. The servo motor, ESP32-CAM and ultrasonic sensor are mounted inside the food container and dispensed.

According to Fig. 4., the food container and dispending unit is used to contain the food and hold the control equipment for the food dispensing unit and has spaces for the ESP32-CAM unit, servo motor and wires. Red color dashed lines show the ESP32-CAM unit partition and the wire path. Magenta color dashed lines show the servo motor partition and its wire path.

Another 3D print part is the food container top cover. This part to the client can open the animal feed machine. It's the food container lid. The amount of food in the container can be measured using the ultrasonic sensor. And it can be remotely detected.

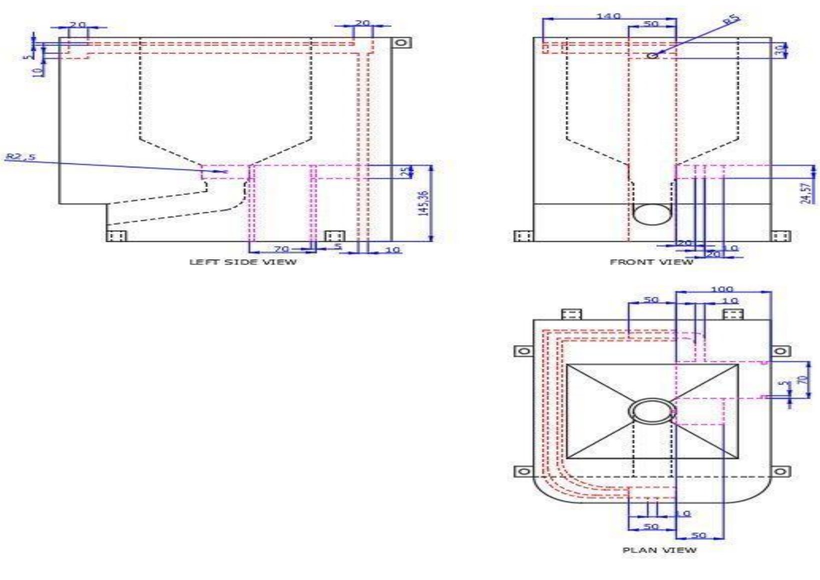


Fig:3.22: Hardware Development

3.6 Software Development

The Firebase is a real-time database that enables secure direct client-side code access, allowing developers to create complex, collaborative apps. Data is locally stored, and real- time events continue to happen even when the user is offline, providing a responsive experience. Fire base is different from conventional app development, which often entails creating both frontend and backend software.

The client does the task in Firebase products since the traditional backend is bypassed. The Firebase console offers access as an administrator to each of these products. Programming on an Arduino board can be made more accessible with the help of the Arduino IDE, an open-source environment. The Arduino IDE is highly user-friendly, and this is likely one of the key reasons why the Arduino platform has become so popular in the first place. The interoperability of a new microcontroller board with the Arduino software development environment is one of the most critical requirements (IDE). The Arduino IDE now allows you to handle third-party boards and libraries while still maintaining the ease of programming the board itself. The Arduino IDE software package includes an IDE and core libraries [9].

The IDE is built in a Java programming environment. C and C++ are the languages of choice for the library core. Field sensor data may be inaccurate or corrupted due to random system failures. Efforts should be made to correct these misinterpretations. During each cycle of ambient data transmission, data packets are generated and processed.

3.6.1 SOFTWARE DESCRIPTION

This project is implemented using following software's:

❖ Proteus simulation– for designing circuit

❖ Arduino compiler - for compilation part

About Proteus

It is a software suite containing schematic, simulation as well as PCB designing.

- **ISIS** is the software used to draw schematics and simulate the circuits in real time.The simulation allows human access during run time,thus providing real time simulation.
- ARES is used for PCB designing.It has the feature of viewing output in 3D view of the designed PCB along with components.
- The designer can also develop 2D drawings for the product.

Features

ISIS has wide range of components in its library. It has sources, signal generators, measurement and analysis tools like oscilloscope, voltmeter, ammeter etc., probes for real time monitoring of the parameters of the circuit, switches, displays, loads like motors and lamps, discrete components like resistors, capacitors, inductors, transformers, digital and analog Integrated circuits, semi-conductor switches, relays, microcontrollers, processors, sensors etc. ARES offers PCB designing up to 14 inner layers, with surface mount and through hole packages. It is embedded with the foot prints of different category of components like ICs, transistors, headers, connectors and other discrete components. It offers Auto routing and manual routing options to the PCB Designer. The schematic drawn in the ISIS can be directly transferred ARES.

3.6.2 Starting New Design

Step 1: Open ISIS software and select New design in File menu

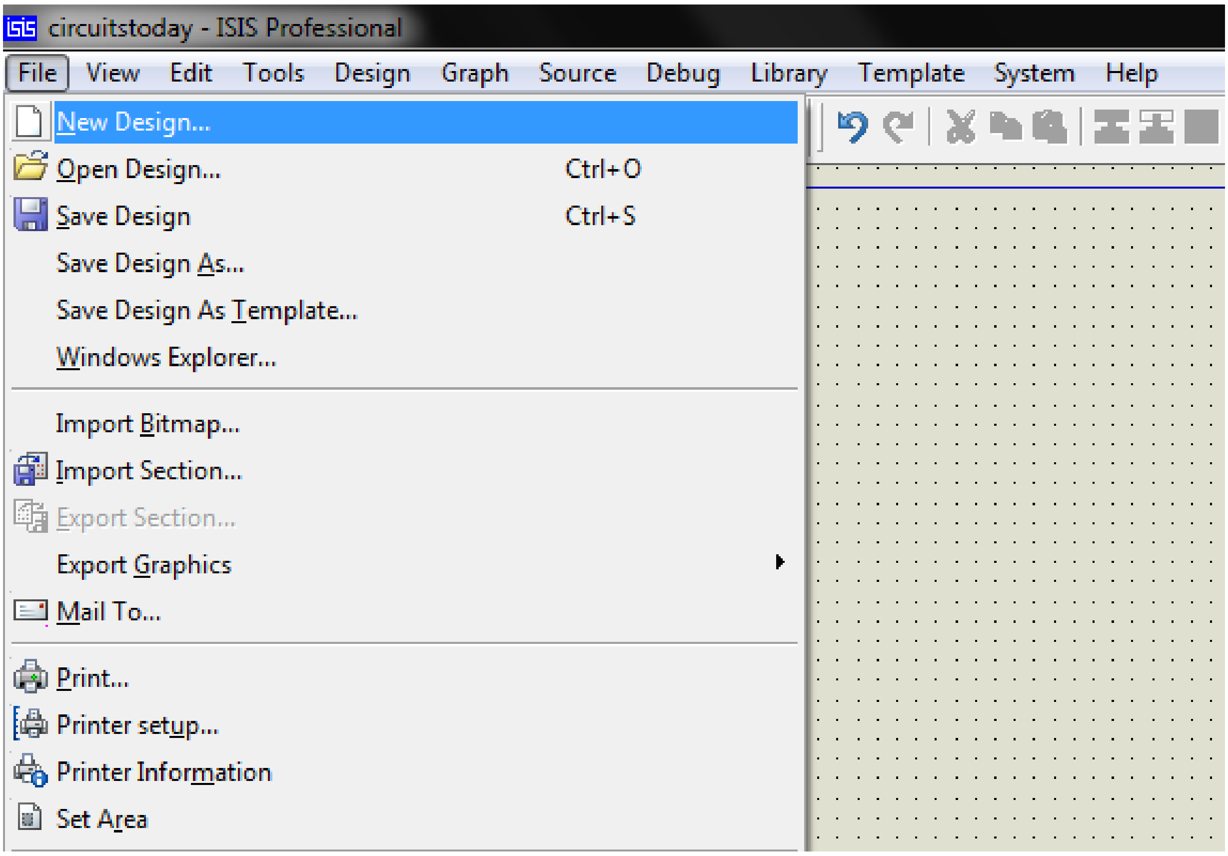


Fig 3.23: Proteus File Menu

Step 2: A dialogue box appears to save the current design. However, we are creating a new design file so you can click Yes or No depending on the content of the present file. Then a Pop-

Up appears asking to select the template. It is similar to selecting the paper size while printing. For now select default or according to the layout size of the circuit.

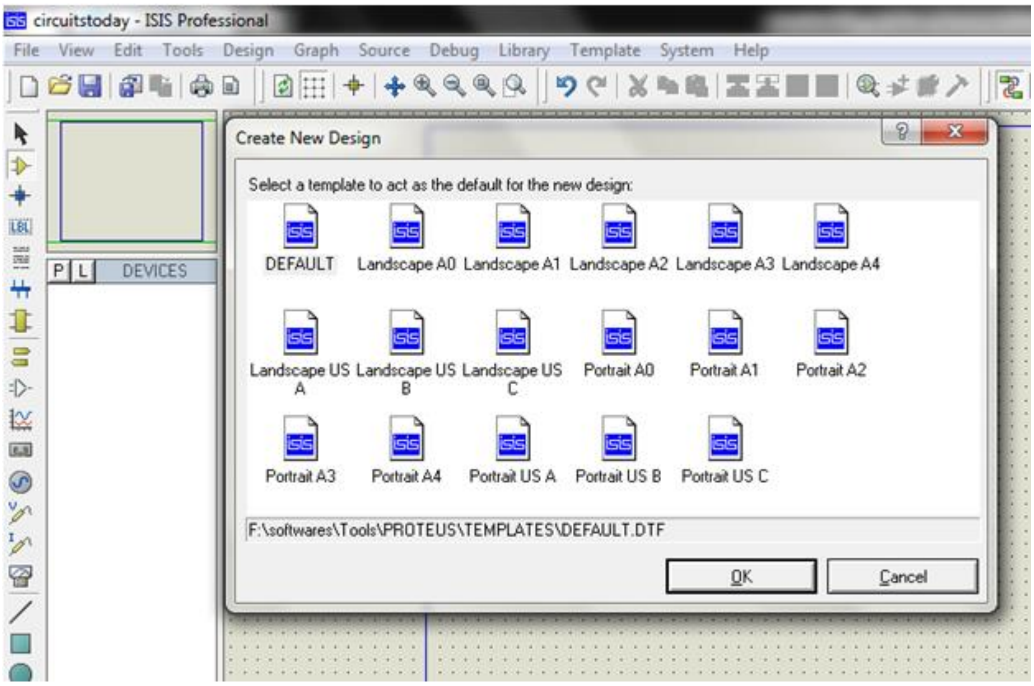


Fig3.24: Proteus Default Template Select

Step 3:An untitled design sheet will be opened, save it according to your wish,it is better to create a new folder for every layout as it generates other files supporting your design. However,it is not mandatory.

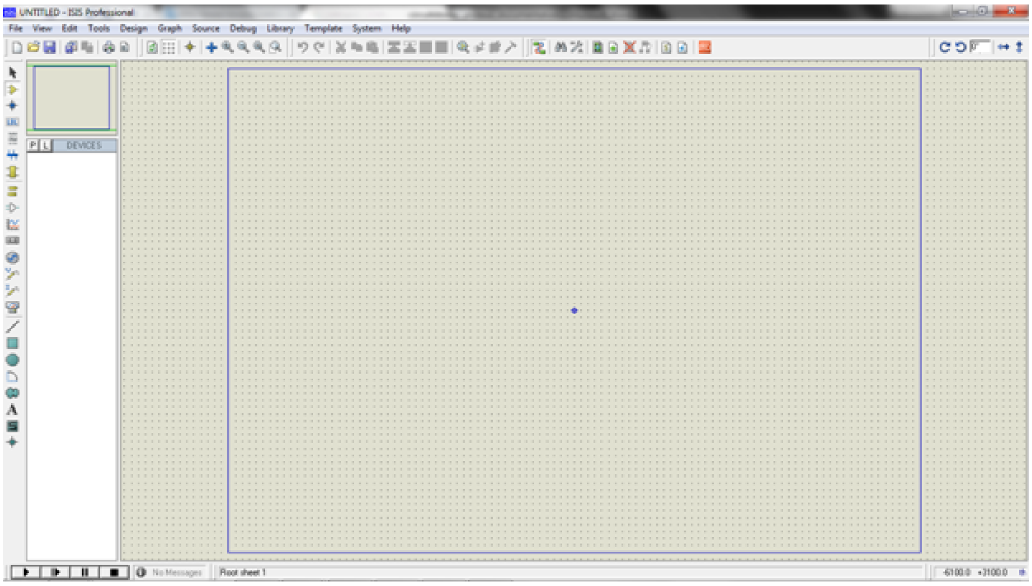


Fig 3.25:Proteus Design Sheet

Step 4:To Select components, Click on the component mode button.

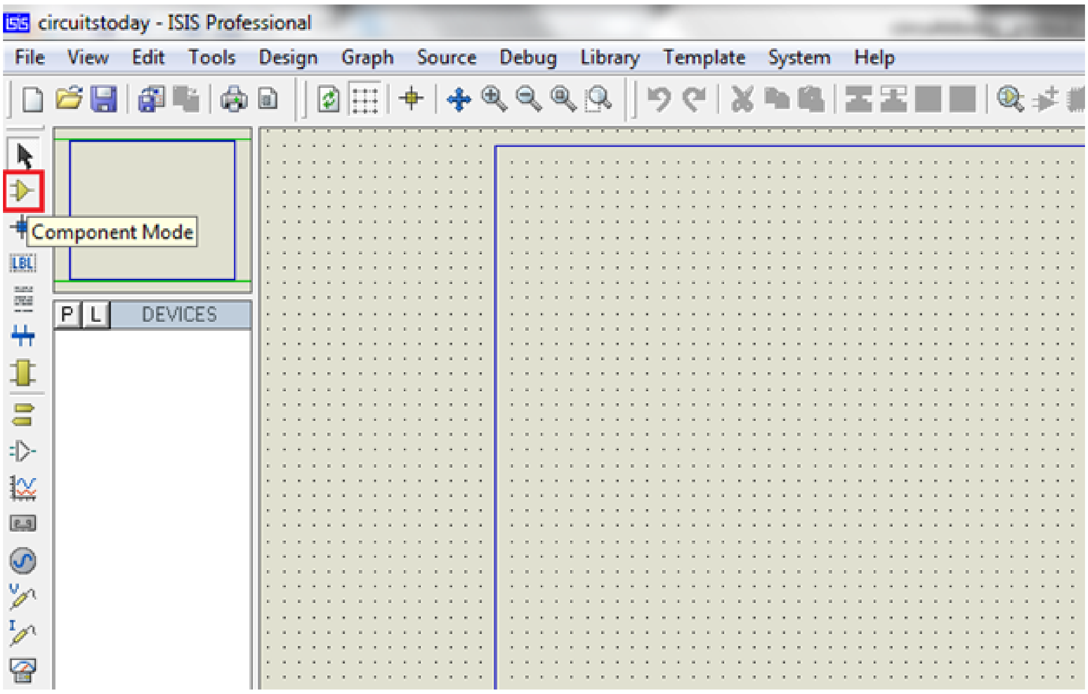


Fig 3.26: Component Mode

Step 5:Click On ARDUINOk from Libraries. It shows the categories of components available and a search option to enter the part name.

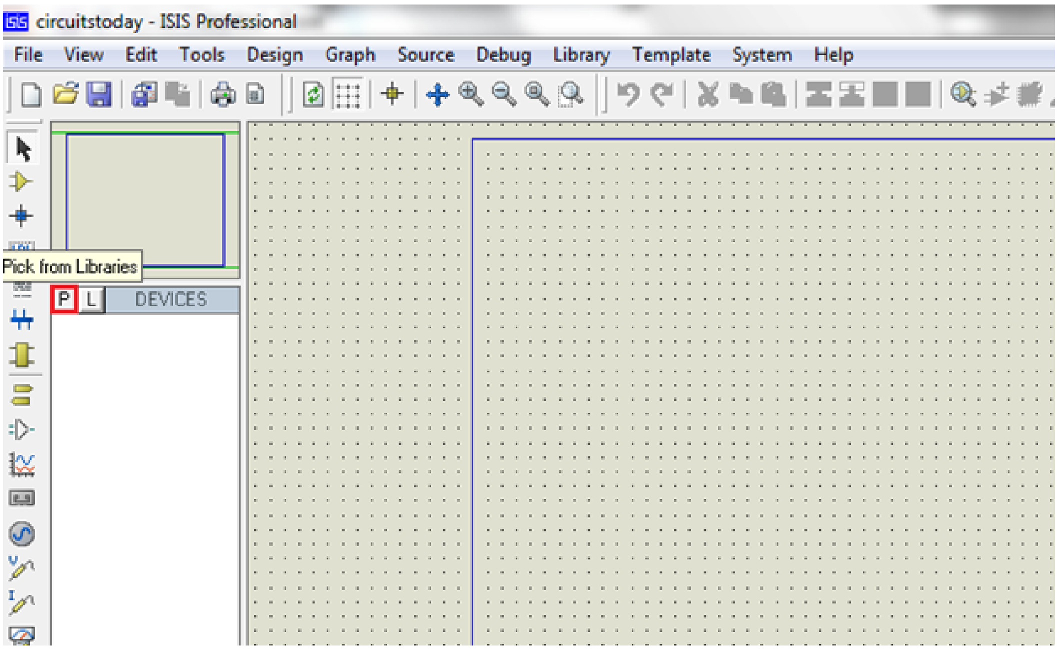


Fig 3.27: ARDUINO k from Libraries

Step 6: Select the components from categories or type the part name in Keywords text box.

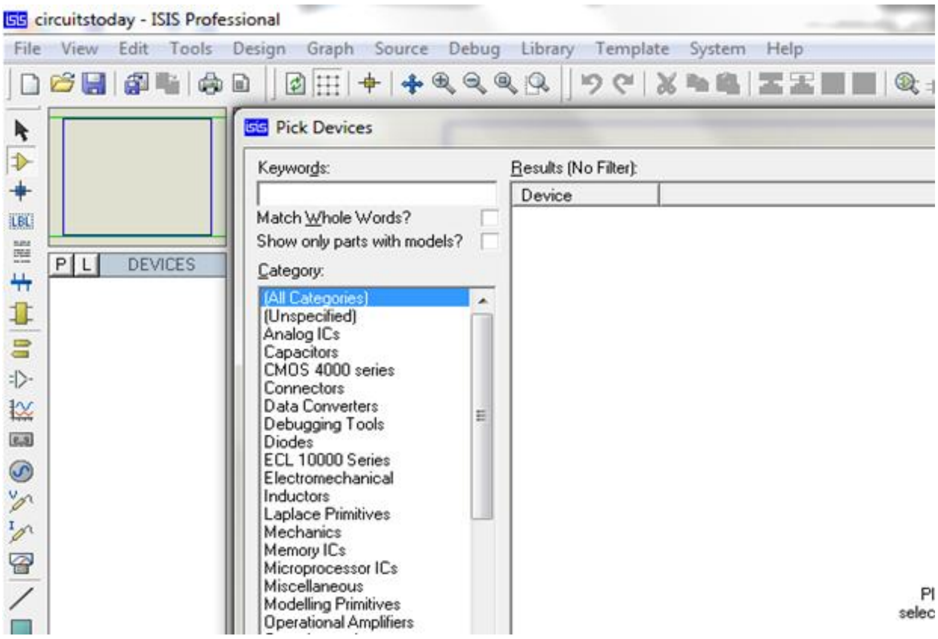


Fig 3.28: Keywords Textbox

Example shows selection of push button. Select the components accordingly.

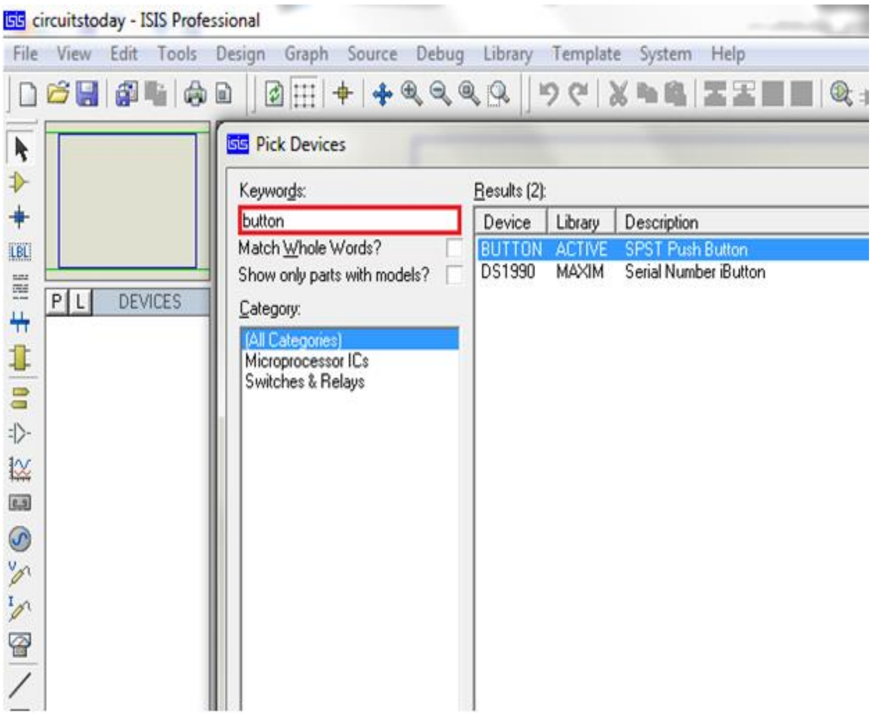


Fig 3.29:Push Button Selection

Step 7: The selected components will appear in the devices list. Select the component and place it in the design sheet by left-click.

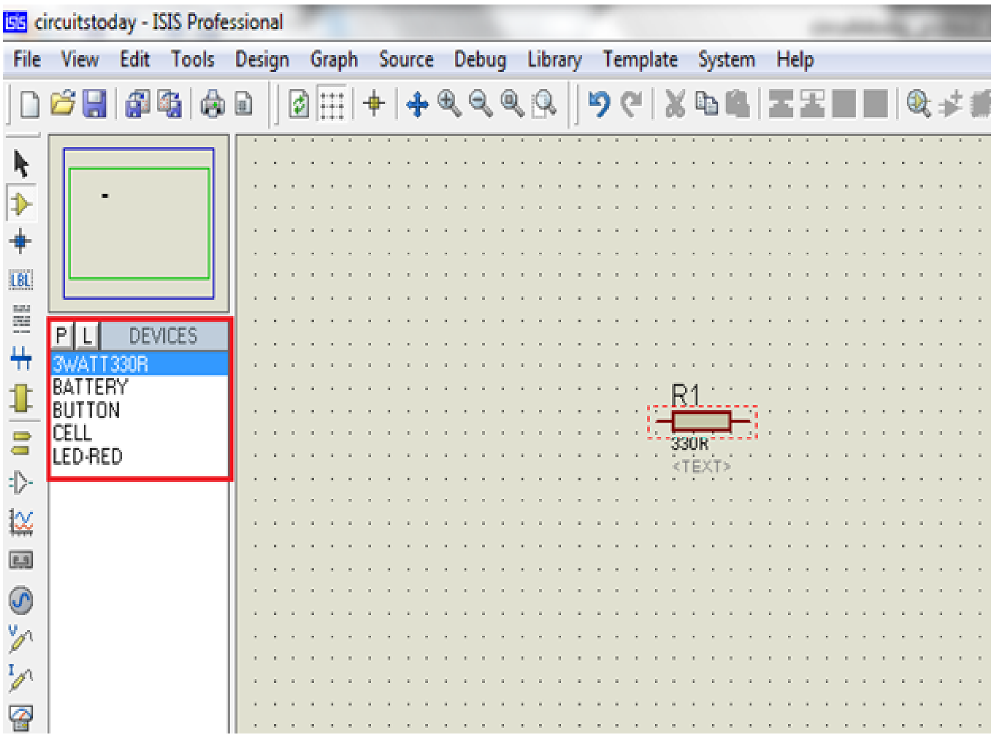


Fig 3.30: Component Selection

Place all the required components and route the wires i.e, make connections.

Either selection mode above the component mode or component mode allows to connect through wires. Left click from one terminal to other to make connection. Double right-click on the connected wire or the component to remove connection or the component respectively.

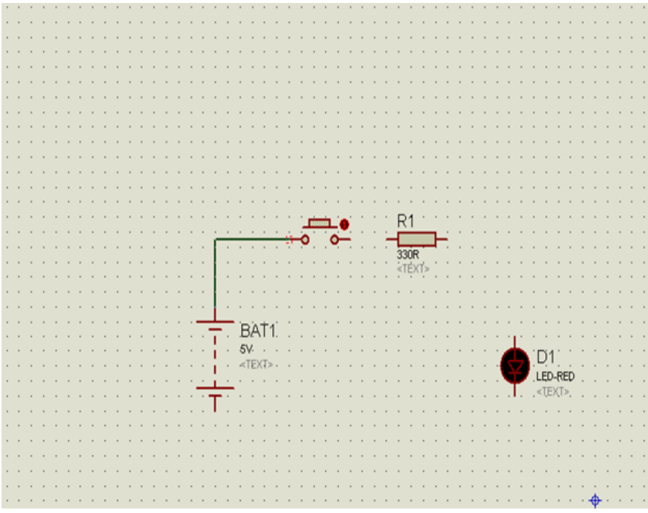


Fig3.31: Component Properties Selection

Double click on the component to edit the properties of the components and click on Ok.

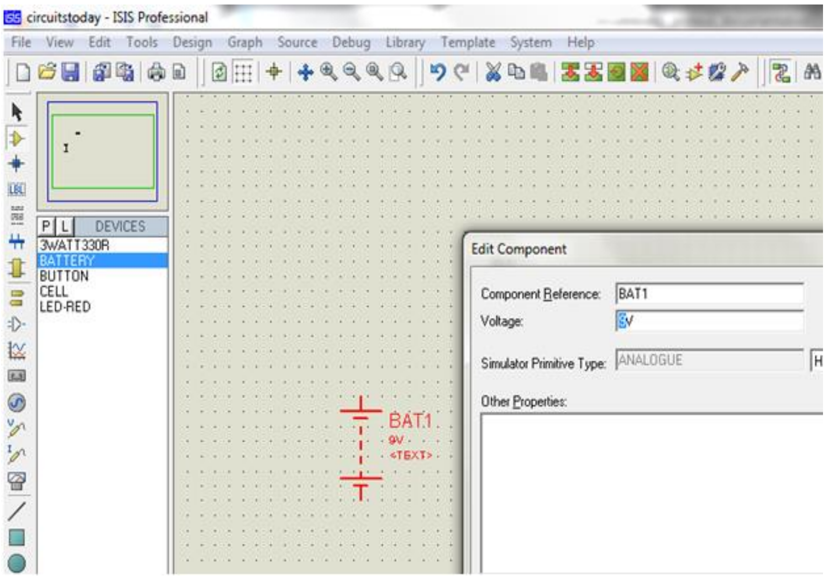


Fig 3.32:Component Properties Edit

Step 8: After connecting the circuit,click on the play button to run the simulation.

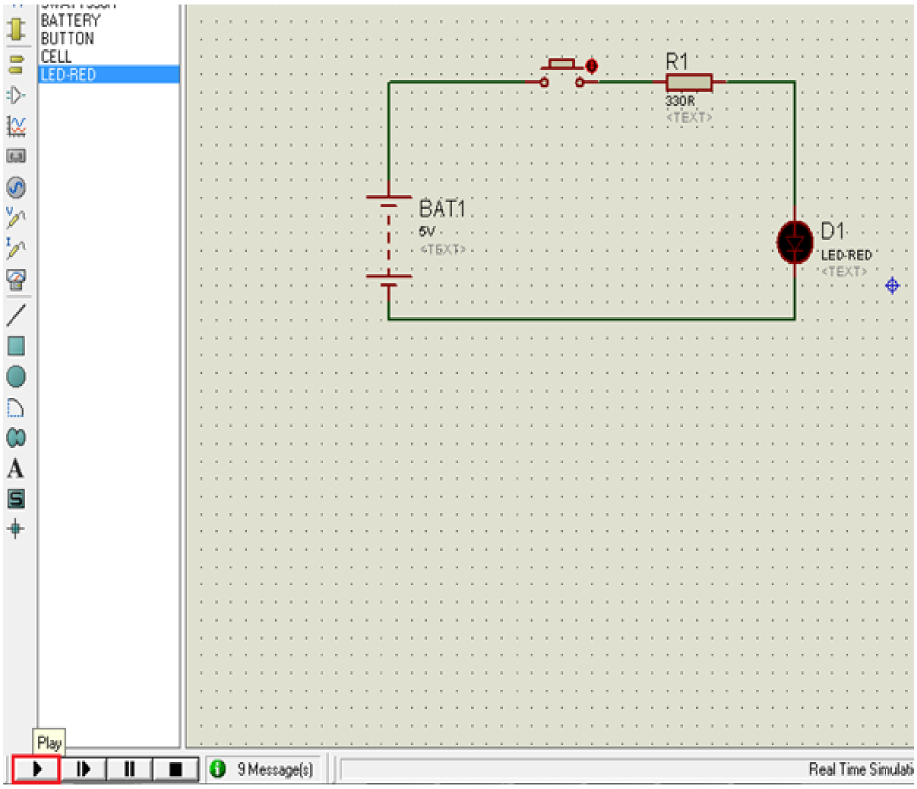


Fig3.33: Simulation Run

In this example simulation, the button is depressed during simulation by clicking on it to make LED glow.

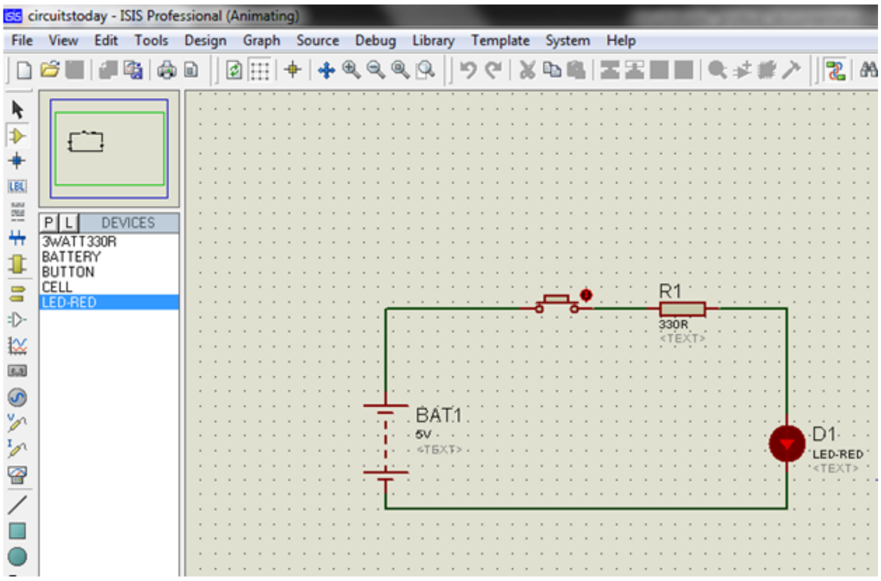


Fig3.34: Simulation Animating

Simulation can be stepped, paused or stopped at any time.

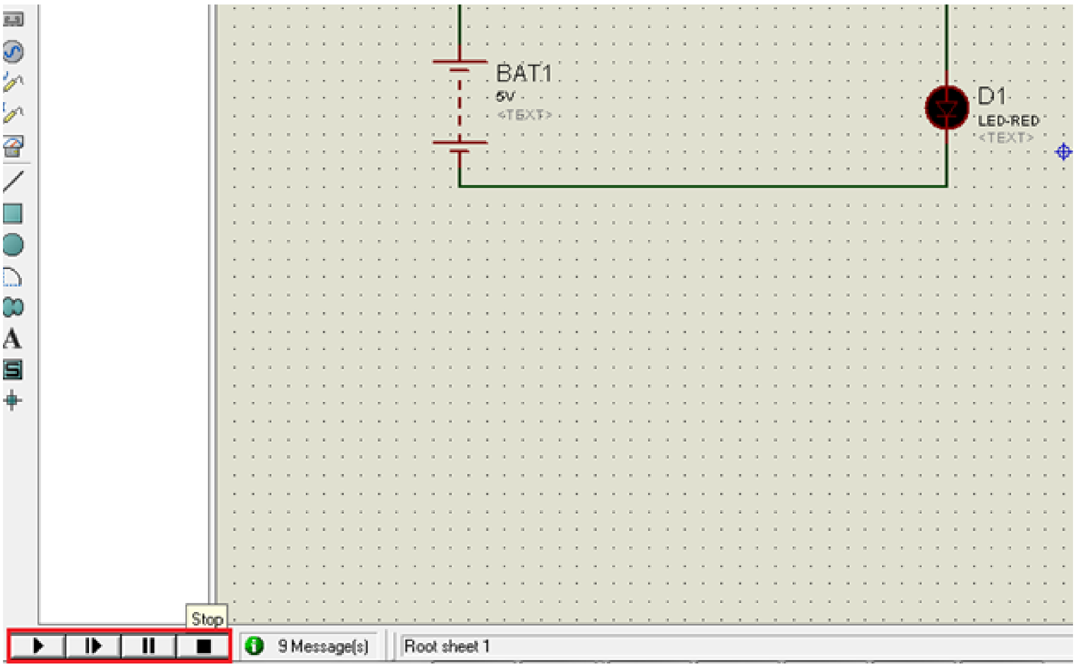


Fig 3.35: Simulation can be stepped

3.7 Simulation Step-Pause-Stop Buttons

Step 1 − First you must have your Arduino board (you can choose your favorite board) and a USB cable. In case you use Arduino UNO, Arduino Duemilanove, Nano, Arduino Mega 2560, or Diecimila, you will need a standard USB cable (A plug to B plug), the kind you would connect to a USB printer as shown in the following image.

Fig 3.36: Simulation Step-Pause-Stop Buttons

In case you use Arduino Nano, you will need an A to Mini-B cable instead as shown in following image.

Step 2 − Download Arduino IDE Software.

You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.

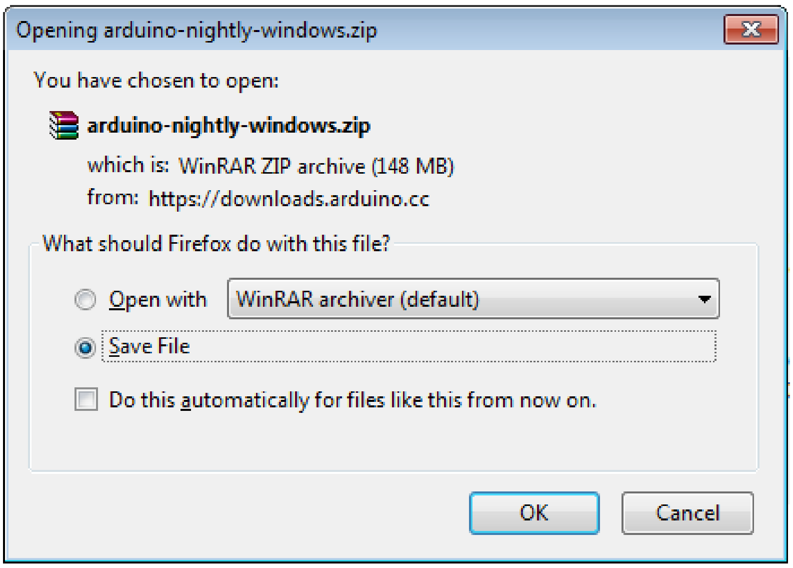


Fig 3.37: Download Arduino IDE Software.

Step 3 − Power up your board.

The Arduino Uno, Mega, Duemilanove and Arduino Nano automatically draw power from either, the USB connection to the computer or an external power supply. If you are using an

Arduino Diecimila, you have to make sure that the board is configured to draw power from the USB connection. The power source is selected with a jumper, a small piece of plastic that fits onto two of the three pins between the USB and power jacks. Check that it is on the two pins closest to the USB port.

Connect the Arduino board to your computer using the USB cable. The green power LED (labeled PWR) should glow.

Step 4 − Launch Arduino IDE.

After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Double-click the icon to start the IDE.

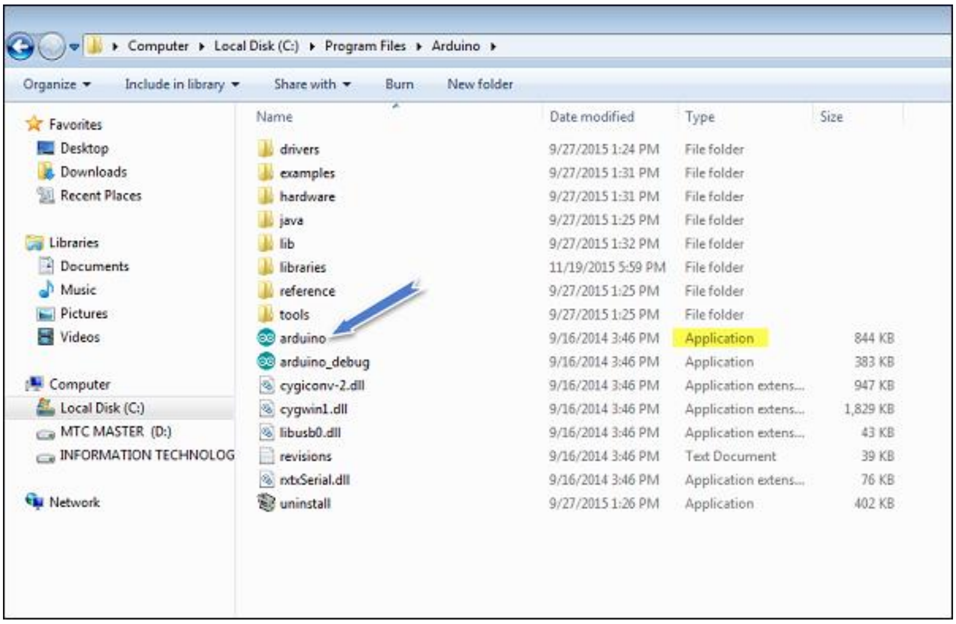


Fig 3.38: Launch Arduino IDE.

Step 5 − Open your first project.

Once the software starts, you have two options −

- Create a new project.
- Open an existing project example.

To create a new project, select File → **New**.

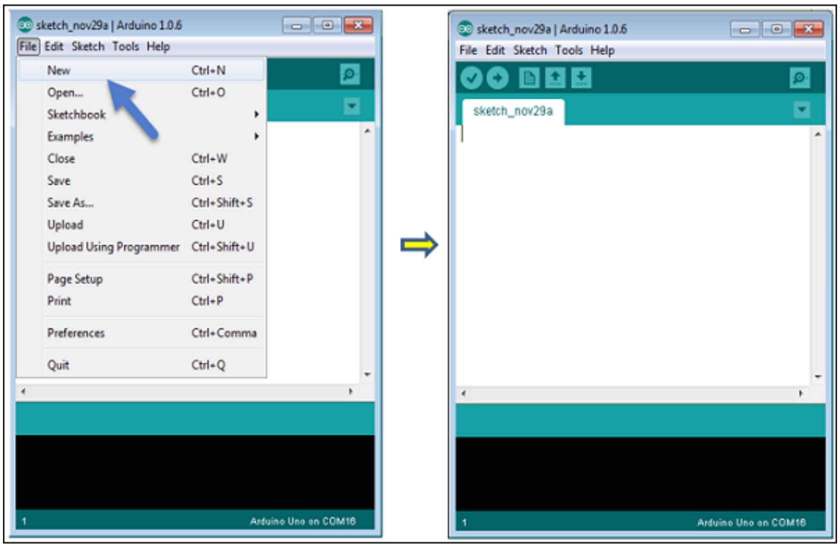


Fig 3.39: Open your first project.

To open an existing project example, select File → Example → Basics → Blink.

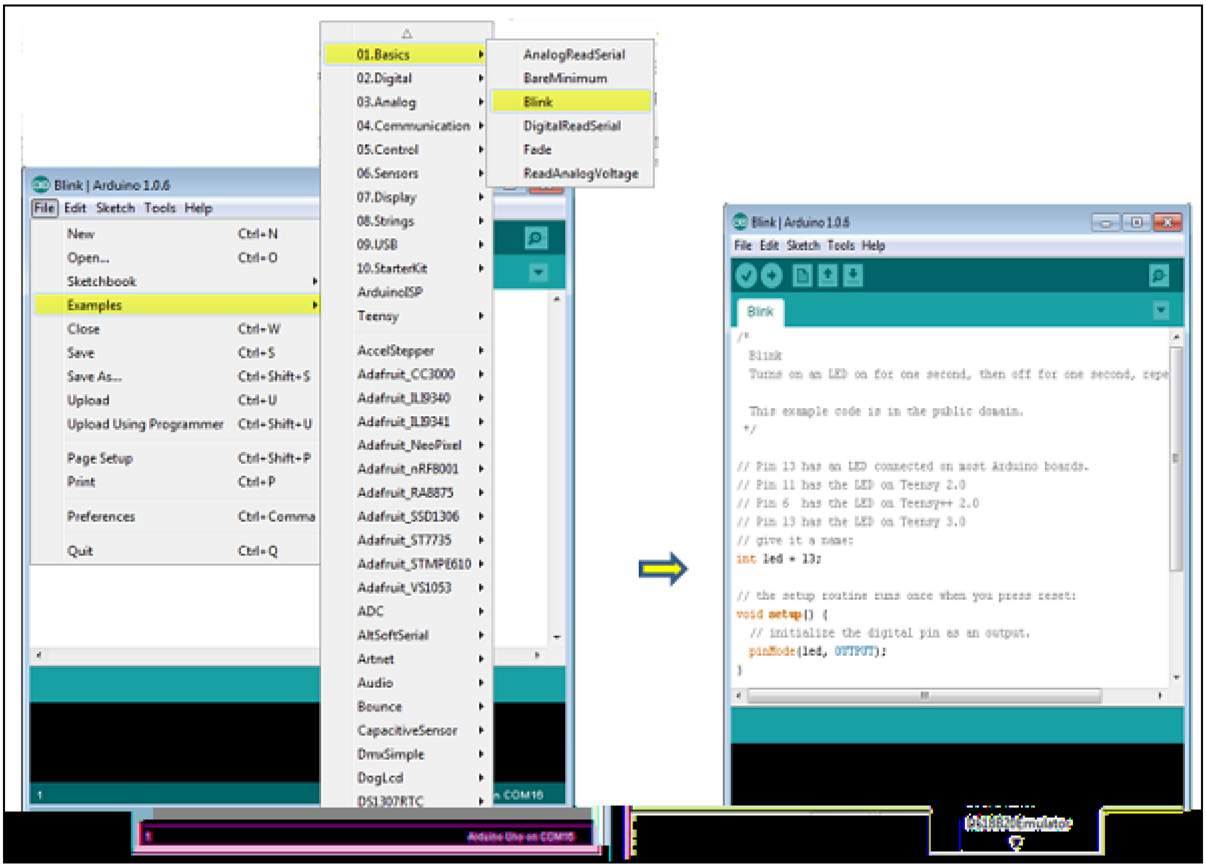


Fig 3.40: Open your first project.

Here, we are selecting just one of the examples with the name **Blink**. It turns the LED on and off with some time delay. You can select any other example from the list.

Step 6 − Select your Arduino board.

To avoid any error while uploading your program to the board, you must select the correct Arduino board name, which matches with the board connected to your computer.

Go to Tools → Board and select your board. [10]

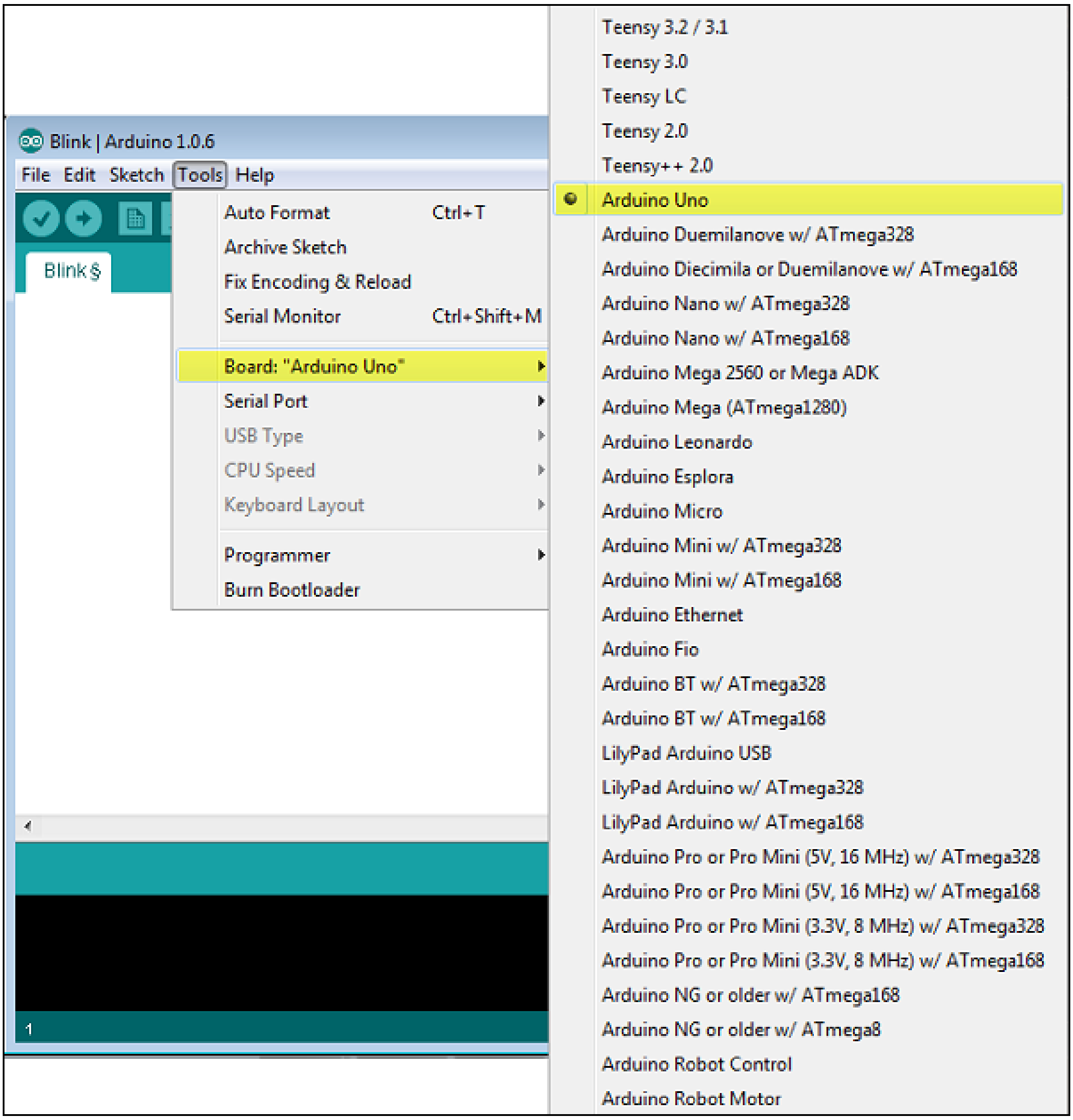


Fig 3.41: Select your Arduino board.

Here, we have selected Arduino Uno board according to our tutorial, but you must select the name matching the board that you are using.

Step 7 − Select your serial port.

Select the serial device of the Arduino board. Go to **Tools → Serial Port** menu. This is likely to be COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports). To find out, you can disconnect your Arduino board and re-open the menu, the entry that disappears should be of the Arduino board. Reconnect the board and select that serial port.

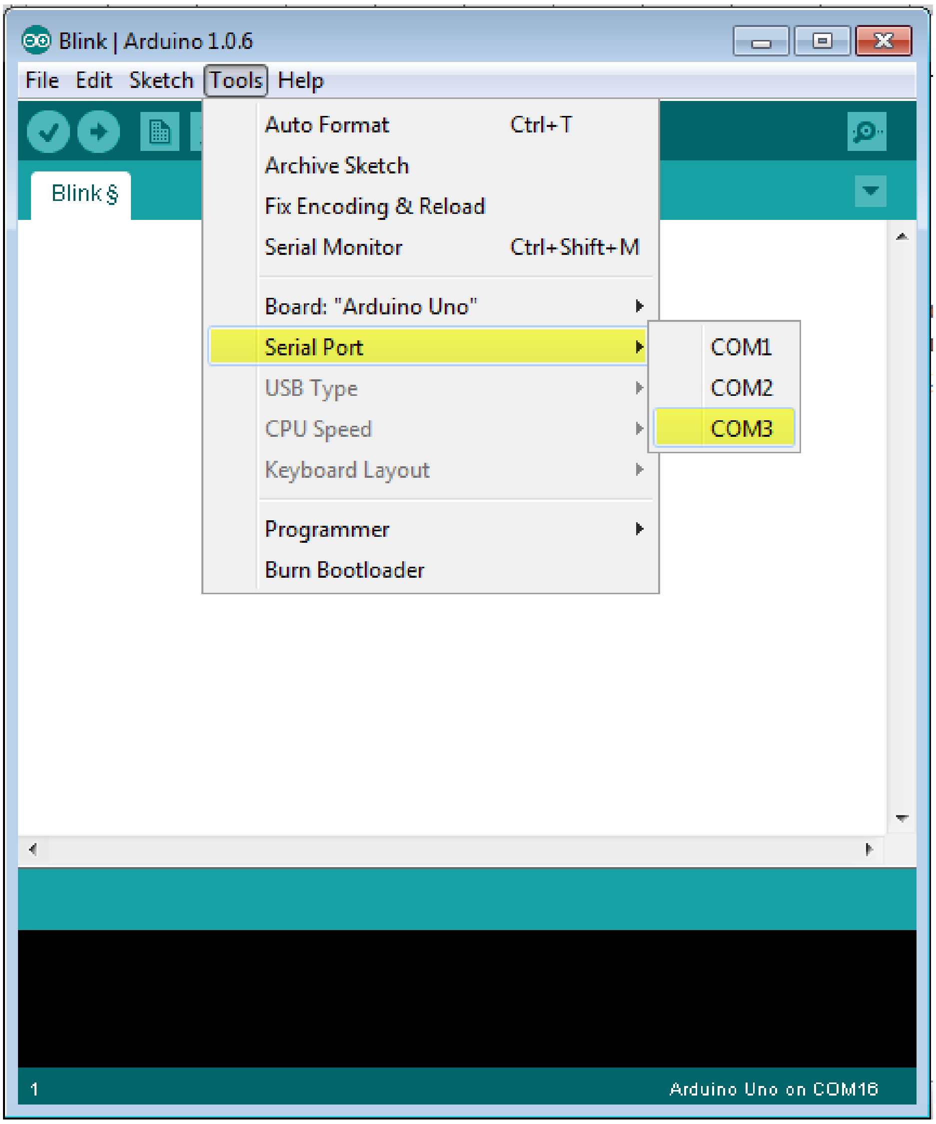


Fig 3.42:Select your serial port.

Step 8 − Upload the program to your board.

Before explaining how we can upload our program to the board, we must demonstrate the function of each symbol appearing in the Arduino IDE toolbar.

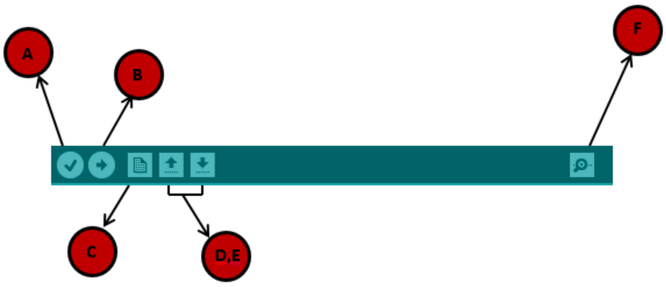


Fig 3.43: Upload the program to your board.

A − Used to check if there is any compilation error.

B − Used to upload a program to the Arduino board.

C − Shortcut used to create a new sketch.

D − Used to directly open one of the example sketch.

E − Used to save your sketch.

F − Serial monitor used to receive serial data from the board and send the serial data to the board.

Now, simply click the "Upload" button in the environment. Wait a few seconds; you will see the RX and TX LEDs on the board, flashing. If the upload is successful, the message "Done uploading" will appear in the status bar.

CHAPTER 4

TESTING AND RESULTS

4.1 INTRODUCTION

Field testing of the finished product was a success, and we got the learning opportunity. The final design of the pet food feeder is shown in Fig. 6. The mechanism used in the pet feeder is rotational to linear motion. The wheels and gear motor assembly are installed between the component's base part. The robot pet food feeder works efficiently and fulfils the objective of feeding pets without its master. The servomotor rotates the propeller, and food gets delivered to the plate according to master mobile control. The whole system is connected to the mobile phone via the internet. The entire real-time data is transmitted to data base and mobile web application.

Fig 4.1: Final Design Of Pet Food Feeder

Also, we develop the client-side web base application for controlling the robot. We used HTML and CSS to create front end of web application. The camara video and robot control keys on the front of web application. [11] [12]

Fig 4.2: Demo Application

CHAPTER 5

CONCLUSION

5.1 CONCLUSION

This paper introduces the design and implementation of pet food feeder robot based on IoT technology. Modern world people have their busy life, and they enjoy with pets. Nowadays, we can see many new devices invented with the aid of IoT. We believe IoT can also change the pattern of the existing structure of pet food feeder robots. In this paper, we describe and implement a new pet food feeder robot that can feed the pets while the owners are absent from their homes and monitor their movement and provide their foods through the owner's smartphones or devices. The implemented system is distinctive from others in that the proposed method is based on IoT, which uses many sensors and wireless communications.

The IoT-based pet food feeder is a helping hand as it works efficiently in the absence of its own. It helps to develop creativity in crating projects and modify existing projects to be more energy efficient with new fabrication methods through this system. This pet food feeder innovation makes it easy for consumers to feed their pets and will not leave their pets hungry again. This automatic pet feeder we developed helps give pets smaller, more frequent meals, which is helpful for weight loss and managing medical disorders such as diabetes. Adequate food intake benefits weight loss and drives medical conditions such as diabetes. When exposed to outside animals, animals or pests cannot damage the remaining food. The future can develop a mobile app for monitoring data and controlling the robot. Another significant development includes the AI-based food care system, which can analyze the behavior of the pet care system for future forecasting.

CHAPTER 6

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